Math 121 Notes

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Contents

1 Metric Spaces

1.1 Open and Closed Sets

Definition 1.1. A metric on a set X is a function $d : X \times X \to \mathbb{R}$ such that the following are true.

(1) For all $x, y \in X$, $d(x, y) = 0$ if and only if $x = y$.

(2) For all $x, y \in X$, $d(x, y) = d(y, x)$.

(3) For all $x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$.

In this case, we say that (X, d) is a metric space.

Remark 1.2. For all $x, y \in X$, $d(x, y) \geq 0$. To see this, note that $0 = d(x, x) \leq d(x, y) + d(y, x) = 2d(x, y)$, so $0 \leq d(x, y)$.

Definition 1.3. The open ball $B_r(x)$ with center $x \in X$ and radius $r > 0$ is defined by

$$B_r(x) = \{y \in X : d(x, y) < r\}.$$

Definition 1.4. A point $y \in Y$, where $Y \subseteq X$ is an interior point of Y if there exists $r > 0$ such that $B_r(y) \subseteq Y$. The collection of interior points is called the interior of Y, and is denoted $\overset{\circ}{Y}$.

Definition 1.5. A subset Y of X is said to be open if ever point of y is an interior point.

Remark 1.6. The empty set $\varnothing$ and the entire space X are open subsets of X.

Proposition 1.7. *An open ball $B_r(x)$ in a metric space X is an open subset of X.*

Proof. Let $y \in B_r(x)$, and let $s = r - d(x, y)$. We show that $B_s(y) \subseteq B_r(x)$. For any $z \in B_s(y)$, the triangle inequality gives

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + s = r + d(x, y) - d(x, y) = r.$$

Thus, $z \in B_r(y)$. ∎

Proposition 1.8. *The interior of a subset $Y \subseteq X$ is an open subset of X.*

Proof. Take $y \in Y$ and take $r > 0$ so that $B_r(y) \subseteq Y$. If $B_r(y) \subseteq \overset{\circ}{Y}$, we are done. If not, there must be some point $y' \in Y \setminus \overset{\circ}{Y} \cap B_r(y)$. Let $s = \min_{y' \in Y \setminus \overset{\circ}{Y} \cap B_r(y)} d(y, y') \geq 0$. Then, if $s \neq 0$, $B_s(y) \subseteq \overset{\circ}{Y}$. If $s = 0$, then for every $\varepsilon > 0$, there is some $y' \in Y \setminus \overset{\circ}{Y} \cap B_\varepsilon(y)$. In particular, for every $r_0 > 0$, there is some $y' \in Y \setminus \overset{\circ}{Y} \cap B_{r_0/2}(y)$, and some $z \in (X \setminus Y) \cap B_{r_0/2}(y')$. Thus, $d(z, y) \leq d(z, y') + d(y', y) < r_0$, meaning that y is not an interior point of Y. ∎

Proposition 1.9. *Let U_i for $i \in I$ be a (possibly infinite) collection of open sets in X. Then $U = \bigcup_{i \in I} U_i$ is also open.*

Proof. Let $x \in U$. Then $x \in U_i$ for some $i \in I$. Since U_i is open, there is some $r > 0$ such that $B_r(x) \subseteq U_i \subseteq U$, so U is open. ■

Proposition 1.10. *A subset U of a metric space is open if and only if U is a union of open balls.*

Proof. If U is the union of open balls, this follows from the previous statements. Suppose that U is open. For each $x \in U$, take $r(x) > 0$ so that $B_{r(x)}(x) \subseteq U$. Then, $U = \bigcup_{x \in U} B_{r(x)}(x)$. ■

Proposition 1.11. *The intersection of any finite number of open sets of a metric space is open.*

Proof. Let $U_1, \ldots, U_n$ be open sets, and let $U = \bigcap_{1 \leq i \leq n} U_i$. Take $x \in U$, and let $r_i > 0$ be such that $B_{r_i}(x) \subseteq U_i$. Let $r = \min_{1 \leq i \leq n} r_i > 0$. Then, $B_r(x) \subseteq B_{r_i}(x) \subseteq U_i$ for all i, so $B_r(x) \subseteq U$. ■

Definition 1.12. A set $Y \subseteq X$ is closed if $X \setminus Y$ is open.

Remark 1.13. The notions of open and closed are not mutually exclusive. Sets can be both or neither. For example, $\varnothing$ and X are both open and closed.

Remark 1.14. Arbitrary intersections of closed sets and finite unions of closed sets are closed, since

$$X \setminus \left(\bigcap_{i \in I} Y_i \right) = \bigcup_{i \in I} (X \setminus Y_i) \qquad \text{and} \qquad X \setminus \left(\bigcup_{i \in I} Y_i \right) = \bigcap_{i \in I} (X \setminus Y_i).$$

Definition 1.15. The closure of $Y \subseteq X$ is the complement of the interior of $X \setminus Y$:

$$\overline{Y} = X \setminus (X \overset{\circ}{\setminus} Y).$$

Remark 1.16. $\overline{Y}$ is closed.

Proposition 1.17. *$Y \subseteq \overline{Y}$.*

Proof. Since $X \overset{\circ}{\setminus} Y \subseteq X \setminus Y$ and $Y \cap (X \setminus Y) = \varnothing$, it follows that $Y \cap (X \overset{\circ}{\setminus} Y) = \varnothing$. Thus, $Y \subseteq X \setminus (X \overset{\circ}{\setminus} Y) = \overline{Y}$. ■

Proposition 1.18. *$y \in \overline{Y}$ if and only if $B_r(y) \cap Y \neq \varnothing$ for all $r > 0$.*

Proof. $y \in \overline{Y}$ if and only if $y \notin X \overset{\circ}{\setminus} Y$. This occurs if and only if, for all $\varepsilon > 0$, $B_\varepsilon(y) \cap Y \neq \varnothing$. ■

Definition 1.19. A point $x \in X$ is said to be adherent to Y if for all $r > 0$, $B_r(x) \cap Y \neq \varnothing$.

Remark 1.20. The closure $\overline{Y}$ consists of all points adherent to Y, which is the set of all points of Y and all points arbitrarily close to Y.

Definition 1.21. Let $\{x_i\}_{i\in\mathbb{N}}$ be a sequence of elements in X. Then $x \in X$ is a limit of the sequence if, for all $r > 0$, there is some N such that $x_n \in B_r(x)$ for all $n > N$. We say that $\{x_i\}_{i\in\mathbb{N}}$ converges to x.

Proposition 1.22. *The limit of a convergent sequence in a metric space is unique.*

Proof. Let $\varepsilon > 0$, and suppose that x, y are limits of the sequence $\{x_i\}_{i\in\mathbb{N}}$. Then, there exists N_1 such that whenever $n \geq N_1$, $d(x_n, x) < \varepsilon/2$, and there exists N_2 such that, whenever $n \geq N_2$, $d(x_n, x) < \varepsilon/2$. By the triangle inequality, $d(x, y) \leq d(x, x_n) + d(x_n, y)$, which is less than ε whenever $n \geq N_1, N_2$. Since ε is arbitrary, $d(x, y) = 0$, meaning that $x = y$. ■

Proposition 1.23. *Let $Y \subseteq X$. A point $x \in X$ is adherent to Y if and only if there is a sequence $\{y_n\}_{n\in\mathbb{N}}$ in Y that converges to X.*

Proof. Suppose that $\{y_n\}_{n\in\mathbb{N}}$ converges to x. Then every open ball centered at x contains terms of $\{y_n\}_{n\in\mathbb{N}}$, meaning that it intersects Y, so x is adherent to Y. Assume that x is adherent to Y. Then for each $n \geq 1$, there is a point $y_n \in B_{\frac{1}{n}}(x) \cap Y$. Then the sequence $\{y_n\}_{n\in\mathbb{N}}$ satisfies that for any $\varepsilon > 0$, whenever $n \geq \frac{1}{\varepsilon}$, we have that $y_n \in B_\varepsilon(x)$, meaning that $\{y_n\}_{n\in\mathbb{N}}$ converges to x. ■

1.2 Completeness

Definition 1.24. A sequence $\{x_n\}_{n\in\mathbb{N}}$ in a metric space X is said to be Cauchy if, for all $\varepsilon > 0$, there is an N such that $d(x_n, x_m) < \varepsilon$ whenever $n, m \geq N$.

Proposition 1.25. *If $\{x_n\}_{n\in\mathbb{N}}$ converges, then $\{x_n\}_{n\in\mathbb{N}}$ is Cauchy.*

Proof. Suppose that $\{x_n\}_{n\in\mathbb{N}}$ converges to x. Then there is an N such that, for $n \geq N$, $d(x_n, x) < \varepsilon/2$. Thus, for $n, m \geq N$, $d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \varepsilon$. ■

Proposition 1.26. *If $\{x_n\}_{n\in\mathbb{N}}$ is a Cauchy sequence and there is a subsequence $\{x_{n_k}\}_{k\in\mathbb{N}}$ that converges to x, then $\{x_n\}_{n\in\mathbb{N}}$ converges to x.*

Proof. Let $\varepsilon > 0$. Since $\{x_{n_k}\}_{k\in\mathbb{N}}$ converges to x, there is some N_1 such that $d(x_{n_k}, x) < \varepsilon/2$ whenever $k \geq N_1$. Since $\{x_n\}_{n\in\mathbb{N}}$ is Cauchy, there is some N_2 such that $d(x_n, x_m) < \varepsilon/2$ whenever $n, m \geq N_2$. Set $N = \max(n_{N_1}, N_2)$. For all $n, n_k \geq N$, we have that $d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon$. ■

Definition 1.27. A metric space X is complete if every Cauchy sequence in X converges (in X).

Remark 1.28. Note every metric space is complete. For example, the interval $(0, 1) \subseteq \mathbb{R}$ with metric $d(x, y) = |x - y|$. The Cauchy sequence $\{\frac{1}{n}\}_{n\in\mathbb{N}}$ does not converge in the interval.

Proposition 1.29. *A closed subspace of a complete metric space is complete.*

Proof. Let $\{y_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence in a closed subspace $Y \subseteq X$. Then $\{y_n\}_{n\in\mathbb{N}}$ is also a Cauchy sequence of X, and thus converges to $x \in X$. Since Y is closed, $x \in Y$. ∎

Proposition 1.30. *A complete subspace Y of a metric space X is closed.*

Proof. Let $\{y_n\}_{n\in\mathbb{N}}$ be a sequence in $Y \subseteq X$ converging to $x \in X$. Then $\{y_n\}_{n\in\mathbb{N}}$ is a convergent sequence of X, and is thus Cauchy. Since Y is complete, $x \in Y$, meaning that Y is closed. ∎

Definition 1.31. A subset $Y \subseteq X$ is said to be dense in X if $\overline{Y} = X$.

Theorem 1.32 (Baire Category Theorem)**.** *If X is a compete metric space and $\{U_i\}_{i\in\mathbb{N}}$ is a sequence of open dense subsets of X, then*

$$U = \bigcap_{i=1}^{\infty} U_i$$

is also dense in X.

Proof. We want to show that for any $x \in X$ and $\varepsilon > 0$, $B_\varepsilon(x) \cap U \neq \varnothing$. We will work interatively. Since U_i is dense in X, there is a $y_i \in B_\varepsilon(x) \cap U_1$. Since U_1 is open, there is some $r_1 > 0$ such that $B_{r_1}(y_1) \subseteq U_1$. By shrinking r_1, we can arrange that $r_1 < 1$, and $\overline{B_{r_1}}(y_1) \subseteq B_\varepsilon(x) \cap U_1$. We repeat this argument with $B_{r_1}(y_1)$ replacing $B_\varepsilon(x)$ and U_2 replacing U_1 to find $y_2 \in B_{r_1}(y_1) \cap U_2$ and $0 < r_2 < 1/2$ such that $\overline{B_{r_2}} \subseteq B_{r_1}(y_1) \cap U_2$. Continuing in this way, we construct a sequence $\{y_i\}_{i\in\mathbb{N}}$ with $y_i \in U_i$ and radii $0 < r_i < 1/i$ such that

$$\overline{B_{r_i}(y_i)} \subseteq B_{r_{i-1}}(y_{i-1}) \cap U_i. \tag{1.1}$$

From this,

$$\overline{B_{r_i}(y_i)} \subseteq \overline{B_{r_{i-1}}(y_{i-1})} \subseteq \cdots \subseteq \overline{B_{r_1}(y_1)} \subseteq B_\varepsilon(x). \tag{1.2}$$

In particular, if $n \geq i$, then $y_n \in B_{r_i}(y_i)$. Since r_i tends to 0 as i approaches ∞, $\{y_i\}_{i\in\mathbb{N}}$ is Cauchy. Since X is complete, this sequence converges to some $y \in X$. Moreover, since $y_n \in B_{r_i}(y_i)$ whenever $n \geq i$, it follows that $y \in \overline{B_{r_i}(y_i)}$. Thus, by (1.1), $y \in U_i$ for all i, and by (1.2), $y \in B_\varepsilon(x)$. Thus, $y \in U \cap B_\varepsilon(x)$. ∎

Definition 1.33. A subset $Y \subseteq X$ is nowhere dense if $\overline{Y}$ has no interior points ($\mathring{\overline{Y}} = \varnothing$).

Remark 1.34. A set Y is nowhere dense if and only if $X \setminus \overline{Y}$ is an open dense set.

Corollary 1.35. *If $\{V_i\}_{i\in\mathbb{N}}$ is a collection of nowhere dense subsets of a complete metric space X, then $\bigcup_{i=1}^{\infty} V_i$ has empty interior.*

Proof. Since the V_i's are nowhere dense, $X \setminus \overline{V_i}$ is open and dense, so $\bigcap_{i=1}^{\infty} X \setminus \overline{V_i} = X \setminus \bigcup_{i=1}^{\infty} V_i$ is dense. Thus, for any $r > 0$ and any element $x \in \bigcup_{i=1}^{\infty} V_i$, $B_r(x) \cap X \setminus \bigcup_{i=1}^{\infty} V_i \neq \varnothing$, so x cannot be an interior point. Thus, $\bigcup_{i=1}^{\infty} V_i$ has an empty interior. ∎

Example 1.36. A single point in $\mathbb{R}$ is nowhere dense, so every countable set in $\mathbb{R}$ has empty interior. Since open sets are their own interior, open sets in $\mathbb{R}$ are either empty or uncountable. Meanwhile, countable sets can be closed ($\mathbb{Z}$), dense ($\mathbb{Q}$), or neither.

1.3 The Real Line

Remark 1.37. $\mathbb{R}$ is a field with the following ordering properties under "$<$".

(1) If $a, b \in \mathbb{R}$, then exactly one of $a < b$, $a = b$ or $b < a$ holds.

(2) If $a < b$ and $b < c$, then $a < c$.

(3) If $a < b$ and $c \in \mathbb{R}$, then $a + c < b + c$.

(4) If $a < b$ and $c > 0$, then $ac < bc$.

This makes $\mathbb{R}$ into a totally ordered field. The rational numbers $\mathbb{Q}$ are also a totally ordered field. The property that characterizes $\mathbb{R}$ up to isomorphism is the Least Upper Bound Axiom.

Definition 1.38. A set $S \subseteq \mathbb{R}$ has a upper bound M if $\forall s \in S$, $s \leq M$. A number M' is a least upper bound if $M' \leq M$ for all upper bounds M.

Definition 1.39. The Least Upper Bound Axiom states that if S is a nonempty subset of $\mathbb{R}$ that is bounded above, then S has a least upper bound.

Theorem 1.40. *The real line $\mathbb{R}$ with the usual metric is a complete metric space.*

Proof. Let $\{x_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence in $\mathbb{R}$ and let $S = \{y : x_n < y \text{ for finitely many } n\}$. If $y \in S$ and $z \leq y$, then $z \in S$. Thus, if $y \in S$, then $(-\infty, y] \subseteq S$. Since $\{x_n\}_{n\in\mathbb{N}}$ is Cauchy, for any $\varepsilon > 0$, there is some N such that for $n, m \geq N$, $d(x_n, x_m) < \varepsilon$. This means in particular that for all $n \geq N$, $x_n \in (x_N - \varepsilon, x_N + \varepsilon)$. Thus, only finitely many of the terms are less than $x_N - \varepsilon$, so that $x_N - \varepsilon \in S$. Moreover, $x_N + \varepsilon$ is a upper bound for S, since if there were some $y \in S$ with $y > x_N + \varepsilon$, then $x + \varepsilon$ would be in S. However, this is impossible, since $x_n < x_N + \varepsilon$ for all $n \geq N$. Thus, S has a least upper bound x. Then, $x_N - \varepsilon \leq x \leq x_N + \varepsilon$, so that $d(x_N, x) < \varepsilon$. Thus, for $n \geq N$, $d(x_n, x) \leq d(x_n, x_N) + d(x_N, x) < \varepsilon + \varepsilon = 2\varepsilon$. Since ε is arbitrary, it follows that $x \in \mathbb{R}$ is the limit of $\{x_n\}_{n\in\mathbb{N}}$. ∎

1.4 Products of Metric Spaces

Definition 1.41. Let $X_1, \ldots, X_n$ be a collection of metric spaces. A product metric on $X = X_1 \times \cdots \times C_n$ is a metric such that the sequence $\{(x_1^i, \ldots, x_n^i)\}_{i\in\mathbb{N}}$ converges to $x = (x_1, \ldots, x_n)$ if and only if for all k, $\{x_k^i\}_{i\in\mathbb{N}}$ converges to x_k.

Remark 1.42. This allows us to say that the open sets in X "look like" the product of open sets in $X_1, \ldots, X_n$.

Theorem 1.43. *Let d be a product metric on $X = X_1 \times \cdots \times X_n$. Then the open sets in (X, d) are unions of products of the form $U_1 \times \cdots \times U_n$, where $U_1 \subseteq X_i$ is open for all i.*

Proof. We first show that if $E_j \subseteq X_j$ is closed, then $E_1 \times \cdots \times E_n$ is closed in X. If $\{x^{(i)}\}_{i\in\mathbb{N}}$ is a sequence in $E_1 \times \cdots \times E_n$ converging to $x = (x_1, \ldots, x_n)$, then the coordinate sequences $\{x_k^{(i)}\}_{i\in\mathbb{N}}$ converge to $x_k \in E_k$. Thus, $x = (x_1, \ldots, x_n) \in E_1 \times \cdots \times E_n$, so the product is closed.

Now we show that products of coordinate open sets are open in X. Suppose that $U_k \subseteq X_k$ is open. Then, the set $X_1 \times \cdots \times X_{k-1} \times (X_k \setminus U_k) \times X_{k+1} \times \cdots \times X_n$ is closed in X. It's complement is $V^{(k)} = X_1 \times \cdots \times X_{k-1} \times U_k \times X_{k+1} \times \cdots \times X_n$ is open in X. Thus the finite intersection $V^{(1)} \cap \cdots \cap V^{(n)} = U_1 \times U_2 \times \cdots \times U_n$ is open in X.

Finally, we show that open sets in X can be expressed as a union of such products. For this, it suffices to show that if $U \subseteq X$ is open and $x \in U$, then there exist open sets $U_k \subseteq X_k$ for all k so that $x \in U_1 \times \cdots \times U_n \subseteq U$. Suppose otherwise that $x \in U$, but $x = (x_1, \ldots, x_n)$ is not contained in any such product. In particular, U does not contain $B_{1/m}(x_1) \times \cdots \times B_{1/m}(x_n)$ for any integer $m > 0$. Thus, this product of balls contains points in $X \setminus U$. Say $x^{(m)} \in X \setminus U$ such that $d_k(x_k^{(m)}, x_k) < 1/m$ for $1 \leq k \leq n$. Then, the sequence $\{x_k^{(m)}\}_{m\in\mathbb{N}}$ converges to x_k. By the definition of a product metric, it follows that $x^{(m)}$ converges to x. Since $X \setminus U$ is closed, $x \in X \setminus U$, which is a contradiction. ■

1.5 Continuity

Definition 1.44. Let X and Y be metric spaces and $f : X \to Y$. We say that f is continuous if, for every open set $U \subseteq Y$, $f^{-1}(U)$ is open in X.

Theorem 1.45. *A map $f : X \to Y$ is continuous if and only if, for all $x \in X$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$.*

Proof. Note that if $U_i \subseteq Y$ is a set of open sets, then $f^{-1}(\bigcup U_i) = \bigcup f^{-1}(U_i)$ and that $f^{-1}(\bigcap U_i) = \bigcap f^{-1}(U_i)$. Since open sets are unions of open balls, $f : X \to Y$ if and only if, for all $y \in Y$ and $\varepsilon > 0$, $f^{-1}(B_\varepsilon(y))$ is open in X. Now $f^{-1}(B_\varepsilon(y))$ is open in X if and only if, for all $x \in f^{-1}(B_\varepsilon(y))$, there is some $\delta > 0$ such that $B_\delta(x) \subseteq f^{-1}(B_\varepsilon(y))$, meaning that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. ■

Theorem 1.46. *A map $f : X \to Y$ is continuous if and only if, for every convergent sequence $\{x_n\}_{n\in\mathbb{N}}$ in x, $\lim_{n\to\infty} f(x_n) = f(\lim_{n\to\infty} x_n)$.*

Proof. Suppose that $f : X \to Y$ is continuous and let $x = \lim_{n\to\infty} x_n$. Since x_n goes to x as n approaches ∞, any δ-ball $B_\delta(x)$ for $\delta > 0$ will contain infinitely many terms of $\{x_n\}_{n\in\mathbb{N}}$. Since f is continuous, for all $\varepsilon > 0$ there is some $\delta > 0$ such that $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$, which is the same as saying that $B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x)))$. Since $B_\delta(x)$ contains infinitely many terms of $\{x_n\}_{n\in\mathbb{N}}$, for all $\varepsilon > 0$, $B_\varepsilon(f(x))$ contains infinitely many terms $\{f(x_n)\}_{n\in\mathbb{N}}$. Thus, $\lim_{n\to\infty} f(x_n) = f(x)$. Now assume that $\lim_{n\to\infty} f(x_n) = f(x)$ and suppose that $f : X \to Y$ is not continuous. Then for some $z \in X$, there is an $\varepsilon > 0$ such that for all $\delta > 0$, there is a z_δ such that $d_X(z_\delta, z) < \delta$ but $d_Y(f(z_\delta), f(z)) \geq \varepsilon$. Then the sequence $\{z_{1/k}\}_{k\in\mathbb{N}}$ approaches z as k approaches ∞, but $d_Y(f(z_{1/k}), f(z)) \geq \varepsilon$. Thus, $\lim_{k\to\infty} f(z_{1/k}) \neq f(z)$, a contradiction. ■

Corollary 1.47. *If B is a dense subset of X and $f : X \to Y$ is continuous, then f is entirely determined by $f|_B$.*

1.6 Compactness

Definition 1.48. A collection of open sets $\{U_i\}_{i\in I}$ in X is called an open cover of $Y \subseteq X$ if $Y \subseteq \bigcup U_i$.

Definition 1.49. A set $Y \subseteq X$ is said to be compact if every open cover $\{U_i\}_{i\in I}$ of Y has a finite subcover of Y. In other words, there is a finite set $\{i_1, \ldots, i_n\} \subseteq I$ such that $Y \subseteq \bigcup_{k=1}^{n} I_{i_k}$.

Proposition 1.50. *If X is compact and $f : X \to Y$ is continuous, then $f(X) \subseteq Y$ is compact.*

Proof. Let $\{U_i\}_{i\in I}$ be an open cover of $f(X)$. Since f is continuous, $\{f^{-1}(U_i)\}_{i\in I}$ is an open cover of X. Thus, there is a finite set $\{i_1, \ldots, i_n\}$ such that $\{f^{-1}(U_{i_k})\}_{1\leq k\leq n}$ is a subcover of X. Then, $\{U_{i_k}\}_{1\leq k\leq n}$ is a finite subcover of $f(X)$. ■

Proposition 1.51. *If $V \subseteq X$ is compact, then V is closed.*

Proof. We show that V^c is open. Let $x \in V^c$. For each $v \in V$, define $\varepsilon_v = \frac{1}{2}d(x, v)$. Note that $B_{\varepsilon_v}(v) \cap B_{\varepsilon_v}(x) = \varnothing$. Now, $\{B_{\varepsilon_v}(v)\}_{v\in V}$ is an open cover of V. Since V is compact, there are $v_1, \ldots, v_n$ such that $\{B_{\varepsilon_{v_i}}(v_i)\}_{1\leq i\leq n}$ is a finite subcover. Let $\varepsilon = \min(\varepsilon_{v_1}, \ldots, \varepsilon_{v_n})$. Then $B_\varepsilon(x) \cap B_{\varepsilon_{v_i}} = \varnothing$, meaning that $B_\varepsilon(x) \cap (\bigcup B_{\varepsilon_{v_i}}(v_i) = \varnothing$, so that in particular, $B_\varepsilon(x) \cap V = \varnothing$. Thus, $B_\varepsilon(x) \subseteq V^c$, so V^c is open. ■

Proposition 1.52. *If X is compact and $V \subseteq X$ is closed, then V is compact.*

Proof. Let $\{U_i\}_{i\in I}$ be an open cover of V. Then $\{U_i\}_{i\in I} \cup \{X \setminus V\}$ is an open cover of X. Since X is compact, there is a finite subcover $\{X \setminus V\} \cup \{U_{i_1}, \ldots, U_{i_n}\}$. It follows that $\{U_{i_k}\}_{1\leq k\leq n}$ is a finite subcover of $\{U_i\}_{i\in I}$ of V. Thus, V is compact. ■

Definition 1.53. A metric space X is totally bounded if, for all each $\varepsilon > 0$, there exists a finite number of points $x_1, \ldots, x_n \in X$ such that $\{B_\varepsilon(x_i)\}_{1\leq i\leq n}$ covers X.

Definition 1.54. A metric space X is bounded if there exists an $r > 0$ such that $d(x, y) < r$ for all $x, y \in X$.

Proposition 1.55. *If X is totally bounded, then X is bounded.*

Proof. Since X is totally bounded, there exist $x_1, \ldots, x_n$ such that $\{B_\varepsilon(x_i)\}_{1\leq i\leq n}$ covers X (for each $\varepsilon > 0$). Let $r' = \max(d(x_i, x_j) : i \neq j)$, and let $r = 2\varepsilon + r'$. Now, given $x, x' \in X$, there are i, j (possible equal) such that $x \in B_\varepsilon(x_i)$ and $x' \in B_\varepsilon(x_j)$. Then,

$$d(x, x') \leq d(x, x_i) + d(x_i, x_j) + d(x_j, x') < \varepsilon + r' + \varepsilon = r.$$

■

Proposition 1.56. *Any subspace of a totally bounded metric space is totally bounded.*

Proof. Let $Y \subseteq X$ be a subspace of a totally bounded metric space X. Let $\varepsilon > 0$, and take $x_1, \ldots, x_n$ so that $\{B_{\varepsilon/2}(x_i)\}_{1\leq i\leq n}$ covers X. Rearranging if necessary, assume that the first k such $\varepsilon/2$-balls intersect Y, and that the last $n - k$ do not. For these first k, chose a point $y_i \in B_{\varepsilon/2}(x_i) \cap Y$. Then, $\{B_\varepsilon(y_i)\}_{1\leq i\leq k}$ is a cover of Y: for $y \in Y$, there is some j such that $y \in B_{\varepsilon/2}(x_j)$, so $d(y, y_j) \leq d(y, x_j) + d(x_j, y_j) < \varepsilon$, so that $y \in B_\varepsilon(y_j)$. ■

Proposition 1.57. *A subset $E \subseteq \mathbb{R}^n$ is totally bounded if and only if it is bounded.*

Proposition 1.58. *Every sequence in a totally bounded metric space X has a Cauchy subsequence.*

Proof. Let $\{x_{1,j}\}_{j\in\mathbb{N}}$ be a sequence in X. We construct a new sequence $\{x_{2,j}\}_{j\in\mathbb{N}}$ as follows. Since X is totally bounded, there are finitely many balls of radius $1/2$ which cover X. At least one of these balls must contain infinitely many terms. Call the first of these terms $x_{2,1}$, the second of these $x_{2,2}$, and so on. We have created a sequence $\{x_{2,j}\}_{j\in\mathbb{N}}$ that is a subsequence of $\{x_{1,j}\}_{j\in\mathbb{N}}$ which is entirely contained in a ball of radius $1/2$. Repeat this process with $1/2$ replaced by $1/3$ and $\{x_{1,j}\}_{j\in\mathbb{N}}$ replaced with $\{x_{2,j}\}_{j\in\mathbb{N}}$ to get a sequence $\{x_{3,j}\}_{j\in\mathbb{N}}$. Continuing, we get a sequence $\{x_{k,j}\}_{j\in\mathbb{N}}$ for each $k \geq 2$ that is a subsequence of $\{x_{k-1,j}\}_{j\in\mathbb{N}}$ and that is entirely contained in a ball of radius $1/k$. Arrange these sequences as follows

$$\begin{matrix} x_{1,1} & x_{1,2} & x_{1,3} & x_{1,4} & \cdots \\ x_{2,1} & x_{2,2} & x_{2,3} & x_{2,4} & \cdots \\ x_{3,1} & x_{3,2} & x_{3,3} & x_{3,4} & \cdots \\ x_{4,1} & x_{4,2} & x_{4,3} & x_{4,4} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{matrix}$$

Take the sequence $\{x_{j,j}\}_{j\in\mathbb{N}}$, and set $y_n = x_{n,n}$. We must have that $\{y_n\}_{n\in\mathbb{N}}$ is a subsequence of $\{x_{1,j}\}_{j\in\mathbb{N}}$, and by construction, $\{y_n\}_{k\leq n<\infty}$ is a subsequence of $\{x_{k,j}\}_{j\in\mathbb{N}}$, which means that $d(y_n, y_m) < 2/k$ whenever $n, m \geq k$. Thus, $\{y_n\}_{n\in\mathbb{N}}$ is Cauchy. ∎

Definition 1.59. A metric space X is separable if there is a dense subset of X that is countable.

Example 1.60. $\mathbb{R}$ is separable, since $\mathbb{Q}$ is countable.

Proposition 1.61. *A subspace of a separable metric space is separable.*

Proof. A countable dense subset of X can be represented by a sequence $\{x_j\}_{j\in\mathbb{N}}$. Let Y be a subspace of X. Since $\{x_j\}_{j\in\mathbb{N}}$ are dense in X, any point of Y is "near" some point of $\{x_j\}_{j\in\mathbb{N}}$, so consider the set S of pairs (j, n) such that $B_{1/n}(x_j) \cap Y \neq \varnothing$. Let $y_{j,n}$ be a point in this intersection. Then $\{y_{j,n}\}_{(j,n)\in S}$ is a countable subset of Y. Let $y \in Y$ and $n \geq 1$. Since $\{x_j\}_{j\in\mathbb{N}}$ is dense in X, there exists a j such that $d(x_j, y) < 1/n$. In particular, $(j, n) \in S$, so we have $y_{j,n}$ such that $d(x_j, y_{j,n}) < 1/n$. Then, $d(y, y_{j,n}) \leq d(y, x_j) + d(x_j, y_{j,n}) < 2/n$. As n approaches ∞, we see that y is adherent to $\bigcup_{(j,n)\in S}\{y_{j,n}\}$. Thus, the closure of $\bigcup_{(j,n)\in S}\{y_{j,n}\}$ (which is countable) is Y, so Y is separable. ∎

Proposition 1.62. *A totally bounded metric space is separable.*

Proof. For each $n \geq 1$, there is a collection of $m(n)$ points $x_{n,1}, \ldots, x_{n,m(n)}$ such that $X = \bigcup_{j=1}^{m(n)} B_{1/n}(x_{n,j})$. The set $S = \{x_{n,j} : 1 \leq n < \infty, 1 \leq j \leq m(n)\}$ is countable. Let $x \in X$ be arbitrary. Now for each n, there is some $x_{n,j}$ such that $x \in B_{1/n}(x_{n,j})$, meaning that $x_{n,j} \in B_{1/n}(x)$. Hence, S is a countable dense subset of X. ∎

Definition 1.63. A collection of open sets is a basis for the topology on X if every open set U is a union of open sets in the basis.

Proposition 1.64. *The set $\mathcal{B} = \{V_i\}_{i\in I}$ is a basis for the topology on X if and only if, for all open sets $U \subseteq X$ and $x \in U$, there is some $V \in \mathcal{B}$ such that $x \in V \subseteq U$.*

Proof. If $\mathcal{B}$ is a basis, then any $U \subseteq X$ is a union of sets in $\mathcal{B}$, so any $x \in U$ is in some $V \in \mathcal{B}$ with $V \subseteq U$. Now suppose that $\mathcal{B}$ has this property. For each $x \in U$, choose some $V_x \in \mathcal{B}$ with $x \in V_x \subseteq U$. Clearly, $U \subseteq \bigcup_{x\in U} V_x \subseteq U$, meaning that $\mathcal{B}$ is a basis. ∎

Definition 1.65. A metric space X is second countable if X has a countable basis.

Theorem 1.66. *A metric space is second countable if and only if it is separable.*

Proof. Suppose that X is second countable. Then there is a sequence $\{U_i\}_{i\in\mathbb{N}}$ which form a basis for X. For each $n \geq 1$, let x_n be a point in U_n. Since $\{U_n\}_{n\in\mathbb{N}}$ is a basis, every open subset contains a term of the sequence $\{x_n\}_{n\in\mathbb{N}}$. Thus, $\bigcup_{n\in\mathbb{N}}\{x_n\}$ is a countable dense subset of X, so X is separable. Now suppose that X is separable. Then there exists a dense sequence $\{x_j\}_{j\in\mathbb{N}}$. Let $\mathcal{B} = \{B_{1/n}(x_j)\ j, n \geq 1\}$ (which is countable). Let U be an open set and take $x \in U$. Since U is open, there is some $N > 0$ such that $B_{1/N}(x) \subseteq U$. Also, since the sequence $\{x_j\}_{j\in\mathbb{N}}$ is dense, there is some $x_k \in B_{1/2N}(x)$. Thus, $x \in B_{1/2N}(x)$. Now, if $z \in B_{1/2N}(x_k)$, then $d(x, z) \leq d(x, x_k) + d(x_k, z) < 1/N$, so $z \in U$. Thus, $\mathcal{B}$ is a basis. ∎

Theorem 1.67 (Lindelöf)**.** *If X is a second countable metric space, then every open cover of X has a countable subcover.*

Proof. Let $\{U_\alpha\}_{\alpha\in A}$ be an open cover of X, and let $\mathcal{B}$ be a countable basis for X. Define $\mathcal{B}' = \{V \in B : \exists\alpha, V \subseteq U_\alpha\}$. We claim that $\mathcal{B}'$ is an open cover of X. If $x \in X$, then for some α, $x \in U_\alpha$, and since $\mathcal{B}$ is a basis, there is a $V \in \mathcal{B}$ such that $x \in V \subseteq U_\alpha$. This implies that $\mathcal{B}'$ covers X. Now for $V \in \mathcal{B}'$, choose $\alpha_V \in A$ such that $V \subseteq U_{\alpha_V}$. Then $\{U_{\alpha_V} : V \in \mathcal{B}'\}$ is a countable open cover of X. ∎

Theorem 1.68. *If X is a metric space, then the following are equivalent:*

(1) X is compact.

(2) Any sequence $\{x_n\}_{n\in\mathbb{N}}$ in X has a convergent subsequence.

(3) X is complete and totally bounded.

Proof. We show that (1) ⇒ (2) ⇒ (3), that (3) ⇒ (2), and that (2)&(3) ⇒ (1).

First we show that (1) ⇒ (2). Let X be compact, and let $\{x_n\}_{n\in\mathbb{N}}$ be a sequence in X. Suppose that for all $x \in X$, there is some $r_x > 0$ such that $B_{r_x}(x)$ contains only finitely many terms of $\{x_n\}_{n\in\mathbb{N}}$. Then, $X = \bigcup_{x\in X} B_{r_x}(x)$. Since X is compact, take $y_1, \ldots, y_k$ so that $X = \bigcup_{1\leq i\leq k} B_{r_{y_i}}(y_i)$. However, this implies that there are only finitely many points of $\{x_n\}_{n\in\mathbb{N}}$ contained in X, a contradiction. Thus, there is some $x \in X$ such that, for all $r > 0$, $B_r(x)$ contains infinitely many terms $\{x_n\}_{n\in\mathbb{N}}$. From this, we create a subsequence $\{x_{n_j}\}_{j\in\mathbb{N}}$ where $x_{n_j} \in B_{1/j}(x)$ and $n_j > n_{j-1}$. By construction, $\{x_{n_j}\}_{j\in\mathbb{N}}$ converges to x.

We now show that (2) ⇒ (3). If $\{x_n\}_{n\in\mathbb{N}}$ is a Cauchy sequence in X with convergent subsequence $\{x_{n_k}\}_{k\in\mathbb{N}}$, then $\{x_n\}_{n\in\mathbb{N}}$ must also converge, as it is Cauchy. Thus, X is complete. Now suppose X is not totally bounded, and take $\varepsilon > 0$ so that there is no finite set of points $x_1, \ldots, x_n$ with $X \subseteq \bigcup B_\varepsilon(x_i)$. Then, for arbitrary $y_1 \in X$, we cannot have that $X \subseteq B_\varepsilon(y_1)$.

Thus, there is some $y_2 \in X \setminus B_\varepsilon(y_1)$. Similarly, we cannot have that $X \subseteq B_\varepsilon(y_1) \cup B_\varepsilon(y_2)$, so there is some $y_3 \in X \setminus (B_\varepsilon(y_1) \cup B_\varepsilon(y_2))$. This process can never terminate, so we have a sequence $\{y_i\}_{i\in\mathbb{N}}$. Note that $d(y_i, y_j) \geq \varepsilon$ for all $i \neq j$. In particular, no subsequence of $\{y_i\}_{i\in\mathbb{N}}$ can converge, a contradiction.

Now we show that (3) $\Rightarrow$ (2). Let X be complete and totally bounded, and let $\{x_n\}_{n\in\mathbb{N}}$ be a sequence in X. Then this has a Cauchy subsequence, as X is totally bounded. Since X is complete, this subsequence converges, so X has a convergent subsequence.

Finally, we show that (2)&(3) $\Rightarrow$ (1). Let $\{U_i\}_{i\in I}$ be an open cover of X. Since X is totally bounded, it is separable. Hence, it has a countable subcover $\{U_j\}_{j\in\mathbb{N}}$. Suppose that $\{U_j\}_{j\in\mathbb{N}}$ has no finite subcover. Then for each n, the set $V_n = X \setminus \bigcup_{j=1}^n U_j$ is nonempty, so choose $x_n \in V_n$. The sequence $\{x_n\}_{n\in\mathbb{N}}$ has a convergent subsequence $\{x_{n_k}\}_{k\in\mathbb{N}}$ with limit x. Since $V_m \subseteq V_n$ whenever $m \geq n$, each V_n contains infinitely many terms of $\{x_k\}_{k\in\mathbb{N}}$. Since V_n is closed, it follows that

$$x \in \bigcap_{i=1}^\infty V_i = \bigcap_{i=1}^\infty (X \setminus \bigcup_{j=1}^i U_j) = X \setminus \bigcup_{j=1}^\infty U_j = \varnothing,$$

a contradiction. Thus $\{U_j\}_{j\in\mathbb{N}}$ has a finite subcover, so X is compact. ∎

Corollary 1.69 (Heine-Borel Theorem)**.** *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. We know that Y is compact if and only if Y is complete and totally bounded. We have seen above that in $\mathbb{R}^n$, Y is totally bounded if and only if Y is bounded. Since $\mathbb{R}^n$ is complete, Y is complete if and only if Y is closed. ∎

2 Topological Spaces

2.1 Topological Spaces

Definition 2.1. A topology on X is a collection $\mathscr{T}$ of subsets of X such that

(1) $\varnothing, X \in \mathscr{T}$.

(2) Any union of sets in $\mathscr{T}$ is also in $\mathscr{T}$.

(3) Any finite intersection of sets in $\mathscr{T}$ is in $\mathscr{T}$.

The sets in $\mathscr{T}$ are called open sets. A topological space is a pair $(X, \mathscr{T})$ (though we will often drop the $\mathscr{T}$ in writing).

Example 2.2. If X is a metric space, then the collection of open sets (as defined by metric balls) is a topology.

Example 2.3. If X is any set, then $\mathscr{T} = \{\varnothing, X\}$ forms a topology called the indiscrete topology.

Example 2.4. If X is any set, the family $\mathscr{T} = \{Y : Y \subseteq X\}$ forms a topology, known as the discrete topology.

Definition 2.5. A topological space $(X, \mathscr{T})$ is called metrizable if there is a metric on X which induces the topology $\mathscr{T}$.

Example 2.6. If $\mathscr{T}$ is the discrete topology on X, then $(X, \mathscr{T})$ is metrizable under the discrete metric. However, no metric on a set with more than one point can induce the indiscrete topology.

Definition 2.7. Let X be a topological space. A set $V \subseteq X$ is closed in X if $X \setminus V$ is open.

Definition 2.8. Let X be a topological space. An open neighborhood of $x \in X$ is an open set U with $x \in U$.

Definition 2.9. Let X be a topological space. A point $x \in X$ is said to be an interior point of $S \subseteq X$ if there exists an open neighborhood U of x such that $U \subseteq S$. The interior of S, denoted $\overset{\circ}{S}$. is the set of all interior points of S.

Proposition 2.10. *The interior of S is open. Also, a set S is open if and only if $S = \overset{\circ}{S}$.*

Proof. If $s \in \overset{\circ}{S}$, there is some open set U_s with $s \in U_s \subseteq S$. In fact, U_s is an open neighborhood of all $s' \in U_s$. Hence, $U_s \subseteq \overset{\circ}{S}$. Thus, $\overset{\circ}{S} \subseteq \bigcup_{s\in \overset{\circ}{S}} U_s \subseteq \overset{\circ}{S}$, meaning that $\overset{\circ}{S} = \bigcup_{s\in\overset{\circ}{S}} U_s$, which is open. Also, if S is open, then S is an open neighborhood of any or its points, so that $S \subseteq \overset{\circ}{S} \subseteq S$, so $S = \overset{\circ}{S}$. ■

Definition 2.11. A point $x \in X$ is said to be adherent to a subset $S \subseteq X$ if every open neighborhood of x intersects S. The closure of S is defined by

$$\overline{S} = X \setminus (X \overset{\circ}{\setminus} S),$$

which consists precisely of all points of X which are adherent to S.

Proposition 2.12. *A subset S of X is closed if and only if $S = \overline{S}$.*

Proof. If $S = \overline{S}$, then $S^c = \overline{S}^c = X \overset{\circ}{\setminus} S$, which is open. If S is closed, then $X \setminus S$ is open, so $X \setminus S = X \overset{\circ}{\setminus} S$, so $\overline{S} = X \setminus (X \overset{\circ}{\setminus} S) = X \setminus (X \setminus S) = S$. ■

Definition 2.13. A sequence of points $\{x_i\}_{i\in\mathbb{N}}$ in a topological space X converges to $x \in X$ if, for every open neighborhood $U \ni x$, there is an integer N such that $x_i \in U$ whenever $i \geq N$.

Proposition 2.14. *If S is a subset of X, and $\{x_i\}_{i\in\mathbb{N}}$ is a sequence in S converging to $x \in X$, then $x \in \overline{S}$.*

Proof. Suppose that $x \notin \overline{S}$. Then $x \in X \setminus \overline{S}$, which is open. Since this is open, there is an open neighborhood U of x which is entirely contained in $X \setminus \overline{S}$. Since $\{x_i\}_{i\in\mathbb{N}}$ converges to x, U contains infinitely many terms of $\{x_i\}_{i\in\mathbb{N}}$. But, $x_i \in S$ for all i, which is a contradiction. ■

Definition 2.15. A point $x \in X$ is a boundary point of $S \subseteq X$ if x is adherent to both S and $X \setminus S$. The boundary of S, denoted ∂S, is the set of all boundary points of S (i.e. $\overline{S} \cap \overline{X \setminus S}$).

Proposition 2.16. *For $S \subseteq X$, $\overline{S}$ is the disjoint union of $\overset{\circ}{S}$ and $\partial(S)$.*

Proof. Clearly, $\overset{\circ}{S} \cup \partial S \subseteq \overline{S}$. Now if $x \in \overline{S}$, then either there is an open neighborhood U of s with $U \subseteq S$, or not. In the former case, $x \in \overset{\circ}{S}$. In the latter case, any open neighborhood of x intersects $X \setminus S$, meaning that $x \in \overline{X \setminus S}$, so that $x \in \partial S$. ■

Definition 2.17. Let $(X, \mathscr{T})$ be a topological space, and let $S \subseteq X$. The family $\mathscr{S} = \{U \cap S : U \in \mathscr{T}\}$ is a topology for S, called the subspace topology or relative topology on S. The elements of $\mathscr{S}$ are called relatively open subsets of S.

2.2 Continuous Functions

Definition 2.18. If X and Y are topological spaces, a map $f : X \to Y$ is said to be continuous if $f^{-1}(V)$ is open, whenever V is an open subset of Y.

Definition 2.19. If X and Y are topological spaces, a map $f : X \to Y$ is continuous at $x \in X$ if, for every open set $V \subseteq Y$ such that $f(x) \in V$, there is some open set $U \subseteq X$ such that $x \in U$ and $f(U) \subseteq V$.

Proposition 2.20. *Let X and Y be be topological spaces. Then, $f : X \to Y$ is continuous if and only if f is continuous at every point of X.*

Proof. Suppose that f is continuous. Take $x \in X$ and let $V \subseteq Y$ be open with $f(x) \in V$. Then $U = f^{-1}(V)$ is an open set in X containing x, and $f^{-1}(U) \subseteq V$, so f is continuous at every point $x \in X$. Now suppose that f is continuous at all $x \in X$. Let V be an open subset of Y, and take $x \in f^{-1}(V)$. Then there is some open set $U_x \subseteq X$ with $x \in U_x$ and $U_x \subseteq f^{-1}(V)$. Define $U = \bigcup_{x \in f^{-1}(V)} U_x$, which is open. Clearly $U \subseteq f^{-1}(V)$, and $f^{-1}(V) \subseteq U$, so $U = f^{-1}(V)$. Thus, $f^{-1}(V)$ is open. ■

Proposition 2.21. *Let X, Y, and Z be topological spaces, and let $f : X \to Y$ and $g : Y \to Z$ be continuous. Then $g \circ f : X \to Z$ is continuous.*

Proof. Let V be an open subset of Z. Then $g^{-1}(V) \subseteq Y$ is an open subset, and so $f^{-1}(g^{-1}(V)) \subseteq X$ is open. Hence, $f^{-1}(g^{-1}(V)) = (g \circ f)^{-1}(V)$ is open. ■

Definition 2.22. A map $f : X \to Y$ is open if, for all open sets $U \subseteq X$, $f(U)$ is open in Y.

Definition 2.23. A map $f : X \to Y$ is a homeomorphism if f is continuous, bijective, and open.

Definition 2.24. Two topological spaces X and Y are homeomorphic if there exists a homeomorphism $f : X \to Y$.

Remark 2.25. If $f : X \to Y$ is a bijection, then f is open if and only if f^{-1} is continuous. Thus, the definition of a homeomorphism can be rephrased as a continuous bijective function with a continuous inverse. An easy consequence of this is that if $f : X \to Y$ is a homeomorphism, then $f^{-1} : Y \to X$ is also a homeomorphism. Note also that compositions of homeomorphisms are homeomorphisms, and that the identity from X to itself is a homeomorphism (keeping the same topology). Thus, "being homeomorphic" is an equivalence relation on the set of topological spaces.

Definition 2.26. A property of a topological space is a topological invariant or "topological property" if it is preserved under homeomorphisms.

2.3 Basis for a Topology

Definition 2.27. A collection $\mathcal{B}$ of open subsets of a topological space X is said to be a basis for the topology of X if every open set U is a union of sets in $\mathcal{B}$.

Proposition 2.28. *The set $\mathcal{B} = \{V_i\}_{i\in I}$ is a basis for the topology on X if and only if, for all open sets $U \subseteq X$ and $x \in U$, there is some $V \in \mathcal{B}$ such that $x \in V \subseteq U$.*

Proof. If $\mathcal{B}$ is a basis, then any $U \subseteq X$ is a union of sets in $\mathcal{B}$, so any $x \in U$ is in some $V \in \mathcal{B}$ with $V \subseteq U$. Now suppose that $\mathcal{B}$ has this property. For each $x \in U$, choose some $V_x \in \mathcal{B}$ with $x \in V_x \subseteq U$. Clearly, $U \subseteq \bigcup_{x\in U} V_x \subseteq U$, meaning that $\mathcal{B}$ is a basis. ■

Proposition 2.29. *A family $\mathcal{B}$ of subsets of X is a basis for a topology if and only if*

(1) Each $x \in X$ lies in at least one set from $\mathcal{B}$

(2) If $U, V \in \mathcal{B}$ and $x \in U \cap V$, then there is some $W \in \mathcal{B}$ with $x \in W \subseteq U \cap V$.

Proof. Assume that $\mathcal{B}$ is a basis. Then $\mathcal{B}$ satisfies (1), as X is open. Moreover, since $U \cap V$ is open, $\mathcal{B}$ must satisfy (2). Now suppose that $\mathcal{B}$ satisfies (1) and (2). Let $\mathscr{T}$ be the family of all the unions of sets from $\mathcal{B}$. It suffices to show that $\mathcal{T}$ is a topology for X (in which case $\mathcal{B}$ is a basis for $\mathscr{T}$). Since $\mathcal{B}$ satisfies (1), clearly $\varnothing, X \in \mathscr{T}$. It is clear that unions of sets in $\mathscr{T}$ are in $\mathscr{T}$. To show that finite intersections of sets in $\mathscr{T}$ are in $\mathscr{T}$, it suffices to show this for any two sets in $\mathscr{T}$. Let $U, V \in \mathscr{T}$, and suppose that $U \cap V \neq \varnothing$ (as in this case, it is clear). Take $x \in U \cap V$. Since U and V are unions of sets in $\mathcal{B}$, there exists U_0 and V_0 with $x \in U_0 \subseteq U$ and $x \in V_0 \subseteq V$, so $x \in U_0 \cap V_0$. Since $\mathcal{B}$ satisfies (2), there is some $W_x \in \mathcal{B}$ such that $x \in W_x$ and $W_x \subseteq U_0 \cap V_0$. Then $W_x \subseteq U \cap V$, and

$$U \cap V = \bigcup_{x \in U\cap V} W_x.$$

Thus, $U \cap V \in \mathscr{T}$. ■

Definition 2.30. A topological space X is second countable if there is a countable basis for the topology of X.

Theorem 2.31 (Lindelöf)**.** *Let X be a second countable topological space. Then every open cover of X has a countable subcover.*

Proof. Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of X, and let $\mathcal{B}$ be a countable basis for X. Define $\mathcal{B}' = \{V \in B : \exists \alpha, V \subseteq U_\alpha\}$. We claim that $\mathcal{B}'$ is an open cover of X. If $x \in X$, then for some α, $x \in U_\alpha$, and since $\mathcal{B}$ is a basis, there is a $V \in \mathcal{B}$ such that $x \in V \subseteq U_\alpha$. This implies that $\mathcal{B}'$ covers X. Now for $V \in \mathcal{B}'$, choose $\alpha_V \in A$ such that $V \subseteq U_{\alpha_V}$. Then $\{U_{\alpha_V} : V \in \mathcal{B}'\}$ is a countable open cover of X. ■

Definition 2.32. A subset S of a topological space is dense in X if $\overline{S} = X$.

Definition 2.33. A topological space X is separable if there is a countable dense subset of X.

Theorem 2.34. *If X is a second countable topological space, then X is separable.*

Proof. Suppose that X is second countable. Then there is a sequence $\{U_i\}_{i \in \mathbb{N}}$ which form a basis for X. For each $n \geq 1$, let x_n be a point in U_n. Since $\{U_n\}_{n \in \mathbb{N}}$ is a basis, every open subset contains a term of the sequence $\{x_n\}_{n \in \mathbb{N}}$. Thus, $\bigcup_{n \in \mathbb{N}} \{x_n\}$ is a countable dense subset of X, so X is separable. ■

2.4 Finite Product Spaces

Definition 2.35. Let $X_1, \ldots, X_n$ be topological spaces. The product topology on the Cartesian product $X = X_1 \times \cdots \times X_n$ is the topology for which a basis is given by the "rectangles"

$$\mathcal{B} = \{U_1 \times \cdots \times U_n : U_i \subseteq X_i \text{ is open}\}.$$

Remark 2.36. The product topology ensures that component projections are continuous. If $\pi_j : X \to X_j$ is the map $\pi_j((x_1, \ldots, x_n)) = x_j$ and $U_j \subseteq X_j$ is open, then $\pi_j^{-1}(U_j) = X_1 \times \cdots \times X_{j-1} \times U_j \times X_{j+1} \times \cdots \times X_n$, which is open in X.

Proposition 2.37. *The product topology $\mathscr{P}$ for $X = X_1 \times \cdots \times X_n$ is the smallest topology for which the component projections π_j are continuous in the sense that, if $\mathscr{T}$ is a topology on X such that the component projections are continuous, then $\mathscr{P} \subseteq \mathscr{T}$.*

Proof. Let $\mathscr{T}$ be a topology for which the projections π_j are $\mathscr{T}$-continuous. Let $U_1, \ldots, U_n$ be open subsets in $X_1, \ldots, X_n$ respectively. We have that $\pi_j^{-1}(U_j) = X_1 \times \cdots \times X_{j-1} \times U_j \times X_{j+1} \times \cdots \times X_n$ is an open set in X, and so

$$\pi_1^{-1}(U_1) \cap \cdots \cap \pi_n^{-1}(U_n) = U_1 \times \cdots \times U_n$$

is open. Thus, $\mathscr{T}$ contains the basis for $\mathscr{P}$, and so $\mathscr{P} \subseteq \mathscr{T}$. ■

Proposition 2.38. *Let $X = X_1 \times \cdots \times X_n$ be the product of topological spaces $X_1, \ldots, X_n$. Then each projection $\pi_j : X \to X_j$ is an open map.*

Proof. Let $U = U_1 \times \cdots \times U_n$ be a basic open set. Then $\pi_j(U) = U_j$ is open. Since maps preserve unions, it follows that the image of any open subset of X under π_j is also open in X_j. ■

Proposition 2.39. *Let E be a topological space, and let f be a map from E to $X = X_1 \times X_n$. Then f is continuous if and only if $(\pi_j \circ f) : E \to X_j$ is continuous for $1 \leq j \leq n$.*

Proof. If f is continuous, since π_j is continuous, $\pi_j \circ f$ is continuous. Now suppose that $\pi_j \circ f$ is continuous for each j. Let $U = U_1 \times \cdots \times U_n$ be a basic open set in X. Then

$$\begin{aligned} f^{-1}(U) &= f^{-1}(\pi_1^{-1}(U_1) \cap \cdots \cap \pi_n^{-1}(U_n)) \\ &= (f^{-1} \circ \pi_1^{-1})(U_1) \cap \cdots \cap (f^{-1} \circ \pi_n^{-1})(U_n) \\ &= (\pi_1 \circ f)^{-1}(U_1) \cap \cdots \cap (\pi_n \circ f)^{-1}(U_n). \end{aligned}$$

Since $\pi_j \circ f$ is continuous for each j, this is the finite intersection of open sets, and so is open. As inverses preserve unions, the preimage of any open set under f is open, so f is continuous. ■

2.5 Quotient Spaces

Definition 2.40. Let X be a set and let $\sim$ be an equivalence relation on X. The equivalence classes of $\sim$ form a partition of X. The set of equivalence classes, denoted $X/\sim$, is called the quotient of X modulo $\sim$. The projection $\pi : X \to X/\sim$ is known as the quotient map.

Definition 2.41. Let $\sim$ be an equivalence relation on a topological space X, and let $\pi : X \to X/\sim$ be the quotient map. The quotient topology on $X/\sim$ is

$$\mathscr{T}_{X/\sim} = \{U \subseteq X/\sim : \pi^{-1}(U) \text{ is open in } X\}.$$

Proposition 2.42. *The quotient topology for $X/\sim$ is the largest topology for which the map $\pi : X \to X/\sim$ is continuous.*

Proof. Suppose that $\mathscr{T}$ is a topology on $X/\sim$ such that π is continuous, and let U be an open set in $X/\sim$. Then $\pi^{-1}(U)$ is open in X, so $U \in \mathscr{T}_{X/\sim}$ by definition. Thus, $\mathscr{T} \subseteq \mathscr{T}_{X/\sim}$. ■

Proposition 2.43. *Let X be a topological space, $\sim$ be an equivalence relation, and $\pi : X \to X/\sim$ be the quotient map. A map f from $X/\sim$ to a topological space Y is continuous if and only if $(f \circ \pi) : X \to Y$ is continuous.*

Proof. Since π is continuous, if f is continuous, then $f \circ \pi$ is continuous. Now suppose that $f \circ \pi$ is continuous. If $V \subseteq Y$ is open, then $(f \circ \pi)^{-1}(V) = \pi^{-1}(f^{-1}(V)) \subseteq X$ is open. Thus, $f^{-1}(V) \subseteq X/\sim$ is open by definition. ■

Proposition 2.44. *Let f be a continuous map from a topological space X to a topological space Y, and let $\sim$ be an equivalence relation on X such that f is constant on each equivalence class of $\sim$. Then there exists a continuous function $g : X/\sim \to Y$ such that $f = g \circ \pi$.*

Proof. We define g on an equivalence class to take the value that f does on that class. By assumption, g is well defined, and $(g \circ \pi) = f$. Since f is continuous, it follows from the previous proposition that g is continuous. ■

2.6 Separation Axioms

Definition 2.45. A topological space X is T_0 if its points are "topologically distinguishable". That is, X is T_0 if, for all $x, y \in X$ with $x \neq y$, there is an open set containing x and not y or containing y and not x.

Definition 2.46. A topological space X is T_1 if, for all $x, y \in X$ with $x \neq y$, there exists an opens set containing x and not y.

Example 2.47. The integers $\mathbb{Z}$ with the cofinite topology (i.e. finite sets are closed) is T_1. Let $n, m \in \mathbb{Z}$. Then, $\mathbb{Z} \setminus \{m\}$ is an open set containing n and not m, and $\mathbb{Z} \setminus \{n\}$ is an open set containing m and not n.

Example 2.48. Let $(\mathbb{R}, \mathscr{T})$ be the topological space where $\mathscr{T} = \{(a, \infty) : a \in \mathbb{R}\}$. Then $(\mathbb{R}, \mathscr{T})$ is T_0, but not T_1. If $a, b \in \mathbb{R}$ with $a < b$, then $(\frac{a+b}{2}, \infty)$ is an open set containing b but not a. However, there is no open set containing a but not b.

Definition 2.49. A topological space X is T_2 (or Hausdorff) if, for all $x, y \in X$ with $x \neq y$, there exist open sets U and V such that $x \in U$, $y \in V$, and $U \cap V = \varnothing$.

Example 2.50. Every metric space is Hausdorff.

Proposition 2.51. *If X is Hausdorff, then limits of sequences in X are unique.*

Proof. Suppose that $\{x_n\}_{n\in\mathbb{N}}$ has limits x and y. Then since X is Hausdorff, there exist open sets $U \ni x$ and $V \ni y$ such that $U \cap V = \varnothing$. However, if $\{x_n\}_{n\in\mathbb{N}}$ approaches x, then there is some N such that $x_n \in U$ for $n \geq N$, and if $\{x_n\}_{n\in\mathbb{N}}$ approaches y, then there is some M such that $x_n \in V$ whenever $n \geq M$. Thus, whenever $x \geq \max(N, M)$, we have that $x_n \in U \cap V = \varnothing$, a contradiction. ∎

Definition 2.52. A topological space X is regular if, for each $x \in X$ and closed subset V not containing x, there exist open sets U_1 and U_2 such that $x \in U_1$, $V \subseteq U_2$ and $U_1 \cap U_2 = \varnothing$.

Remark 2.53. A regular space is not Hausdorff unless singleton sets $\{x\}$ are closed.

Definition 2.54. A topological space X is T_3 if it is regular and T_1.

Remark 2.55. A T_3 space is Hausdorff. To see this, let X be T_3, and let $x \neq y \in X$. Then, there is an open set U containing x and not y. Then U^c is closed and does not contain x, so there are open sets V and W such that $x \in V$ and $U^c \subseteq W$ with $V \cap W = \varnothing$. Since $y \in U^c \subseteq W$, we have that there are open sets V and W such that $x \in V$, $y \in W$, and $V \cap W = \varnothing$, so X is Hausdorff.

Definition 2.56. A topological space X is normal if, for all closed V_1, V_2 with $V_1 \cap V_2 = \varnothing$, there are open sets U_1, U_2 with $V_1 \subseteq U_1$, $V_2 \subseteq U_2$ and $U_1 \cap U_2 = \varnothing$.

Definition 2.57. A topological space X is T_4 if it is normal and T_1.

Remark 2.58. If X is T_4, then X is T_3. Let X be a point and V be a closed set not containing x. If V is empty, then this is trivially satisfied, so suppose that V is nonempty. Take $y \in V$. Then, as X is T_1, there is an open set U containing y but not x. Since this is true for all $y \in V$, we may take U so that $V \subseteq U$. Then, U^c is a closed set that is disjoint from V, so there are open sets W and Z such that $U^c \subseteq W$, $V \subseteq Z$, and $W \cap Z = \varnothing$. In particular, $x \in W$, $V \subseteq Z$, and $W \cap Z = \varnothing$, so X is regular, and thus T_3.

Theorem 2.59. *Every metric space is T_4.*

Proof. Let E and F be disjoint closed subsets of a metric space. For each $x \in E$, there is a $r_x > 0$ such that $B_{r_x}(x) \cap F = \varnothing$. Similarly, for each $y \in F$, there is $r_y > 0$ such that $B_{r_y}(y) \cap E = \varnothing$. It follows that

$$E \subseteq \bigcup_{x \in E} B_{r_x/2}(x) := U$$

and that

$$F \subseteq \bigcup_{y \in F} B_{r_y/2}(y) := V.$$

Note that U and V are open. Now suppose that $U \cap V \neq \varnothing$. Then there are $x \in E$, $y \in F$, and $z \in U \cap V$ such that $z \in B_{r_x/2}(x) \cap B_{r_y/2}(y)$. Then, $d(x,y) \leq d(x,z) + d(z,y) < r_x/2 + r_y/2 \leq \max(r_x, r_y)$. However, if $r_x \geq r_y$, then since $d(x,y) < r_x$, we have that $y \in B_{r_x}(x)$, a contradiction. Similarly if $r_y \geq r_x$, then $x \in B_{r_y}(y)$. This cannot happen, so $U \cap V = \varnothing$. ■

Lemma 2.60. *A topological space X is normal if and only if, for each closed set V and open set U with $V \subseteq U$, there is an open set W such that*

$$V \subseteq W \subseteq \overline{W} \subseteq U.$$

Proof. Suppose that X is normal, and that V is a closed set contained in the open set U. Then V and $X \setminus U$ are disjoint closed sets. Since X is normal, there are disjoint open sets W and W' such that $V \subseteq W$, $X \setminus U \subseteq W'$, and $W \cap W' = \varnothing$. Since $W \cap W' = \varnothing$, $W \subseteq X \setminus W'$. Since W' is open, $X \setminus W'$ is closed, so $\overline{W} \subseteq X \setminus W'$. As $X \setminus U \subseteq W'$, $X \setminus W' \subseteq U$, so that $V \subseteq W \subseteq \overline{W} \subseteq U$. Now suppose that X has this property. Let V_1 and V_2 be disjoint closed sets. Then $V_1 \subseteq X \setminus V_1$, which is open. Thus, there is an open set U_1 such that

$$V_1 \subseteq U_1 \subseteq \overline{U_1} \subseteq X \setminus V_2.$$

Write $U_2 = X \setminus \overline{U_1}$, which is open. Then, as $\overline{U_1} \subseteq X \setminus V_2$ $V_2 \subseteq X \setminus \overline{U_1} = U_2$. Thus, $V_1 \subseteq U_1$ and $V_2 \subseteq U_2$. Since $U_1 \subseteq \overline{U_1} = X \setminus U_2$, $U_1 \cap U_2 = \varnothing$, so X is normal. ■

Definition 2.61. The dyadic rationals are the set

$$\mathcal{D} = \left\{ \frac{a}{2^n} : a, n \in \mathbb{Z} \right\}.$$

Proposition 2.62. *The dyadic rationals are dense in $\mathbb{R}$.*

Proof. Take $(a, b) \subseteq \mathbb{R}$, and choose n such that $2^{-n} < b - a$. Then, $2^n b - 2^n a > 1$, so there must be an integer m with $2^n a < m < 2^n b$. Thus, $a < m/2^n < b$, so there is a dyadic rational in (a, b). ■

Theorem 2.63 (Urysohn's Lemma)**.** *If X is a normal topological space, and E and F are disjoint closed subsets, then there exists a continuous function $f : X \to [0, 1]$ such that $f = 0$ on E and $f = 1$ on F.*

Proof. The set $V = X \setminus F$ is an open set containing E, so by the previous lemma, there is $U_{1/2}$ such that $E \subseteq U_{1/2} \subseteq \overline{U_{1/2}} \subseteq V$. We repeat this process with $E \subseteq U_{1/2}$ and with $\overline{U_{1/2}} \subseteq V$ to find $U_{1/4}$ and $U_{3/4}$ such that

$$E \subseteq U_{1/4} \subseteq \overline{U_{1/4}} \subseteq U_{1/2} \subseteq \overline{U_{1/2}} \subseteq U_{3/4} \subseteq \overline{U_{3/4}} \subseteq V.$$

Continuing this way, we create an open subset U_r for each dyadic rational $r \in (0, 1)$ with $E \subseteq U_r \subseteq V$, and $\overline{U_r} \subseteq U_s$ for $0 < r < s < 1$. Now define $f : X \to [0, 1]$ by

$$f(x) = \begin{cases} 0 & x \in U_r \forall r \in (0,1) \cap \mathcal{D} \\ \sup\{r : x \notin U_r\} & \text{else} \end{cases}.$$

By construction, $f = 0$ on E and $f = 1$ on F. We must show that f is continuous. Let $x \in X$, and assume $0 < f(x) < 1$. For any $\varepsilon > 0$, there is $r, s \in (0, 1) \cap \mathcal{D}$ such that

$$f(x) - \varepsilon < r < f(x) < s < f(x) + \varepsilon.$$

By construction of f, $x \notin U_t$ for any dyadic rational $t \in (r, f(x))$. Since

$$U_r \subseteq \overline{U_r} \subseteq U_t,$$

we have that $x \notin U_r$. We also have that $x \in U_s$, as $f(x) < s$. Thus, $U_s \setminus \overline{U_r}$ is an open neighborhood of x, and for $y \in U_s \setminus \overline{U_r}$, we have that

$$f(x) - \varepsilon < r \leq f(y) \leq s < f(x) + \varepsilon,$$

so that $f(y) \in B_\varepsilon(f(x))$. Now suppose that $f(x) = 0$. There is a dyadic rational $0 = f(x) < s < \varepsilon$. By definition, $x \in U_s$, so U_s is an open neighborhood of x. Take $y \in U_s$. Then, $f(y) \leq s < \varepsilon$, so $f(y) \in B_\varepsilon(f(x))$. Finally, suppose that $f(x) = 1$. There is a dyadic rational $1 - \varepsilon < s < 1 = f(x)$. There must be some dyadic rational $s < t < f(x)$ such that $x \notin U_t$, so that $X \setminus \overline{U_t}$ is an open neighborhood of x. For $y \in X \setminus \overline{U_t}$, we have that $\varepsilon < s < t \leq f(y)$, so $f(y) \in B_\varepsilon(f(x))$. ■

Definition 2.64. Let S be a set and X be a metric space. Define d_∞ on $F(S, X) = \{f : S \to X\}$ by $d_\infty(f, g) = \sup_{s \in S}\{\min(1, d(f(s), g(s))\}$.

Proposition 2.65. *The function d_∞ is a metric on $F(S, X)$.*

Proof. Clearly, $d_\infty(f,f) = 0$ and $d_\infty(f,g) = d_\infty(g,f)$. Now,

$$\begin{aligned}
d_\infty(f,g) &= \sup_{s\in S}\{\min(1, d(f(s)g(s)))\} \leq \sup_{s\in S}\{\min(1, d(f(s),h(s)) + d(h(s),g(s))\} \\
&\leq \sup_{s\in S}\{\min(1, d(f(s),h(s))) + \min(1, d(h(s),g(s)))\} \\
&\leq \sup_{s\in S}\{\min(1, d(f(s),h(s)))\} + \sup_{s\in S}\{\min(1, d(h(s),g(s)))\} \\
&= d_\infty(f(s),h(s)) + d_\infty(h(s),g(s)).
\end{aligned}$$

■

Definition 2.66. A sequence of functions $\{f_n\}_{n\in\mathbb{N}}$ converges uniformly to f if, for each $\varepsilon > 0$, there is some N such that $d(f_n(s), f(s)) < \varepsilon$ for all $s \in S$, whenever $n > N$.

Proposition 2.67. *Convergence of a sequence of functions in the metric space $(F(S,X), d_\infty)$ is equivalent to uniform convergence.*

Proof. Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of functions converging to f, and take $0 < \varepsilon < 1$. Then, there is some N such that, for $n > N$, $d_\infty(f_n, f) < \varepsilon$ for $n > N$. Thus, $\sup_{s\in S}\{\min(1, d(f_n(s), f(s)))\} < \varepsilon$. Since $1 \not< \varepsilon$, we have that $\sup_{s\in S}\{d(f_n(s), f(s))\} < \varepsilon$. Thus, for $n > N$, $d(f_n(s), f(s)) < \varepsilon$, so that $\{f_n\}_{n\in\mathbb{N}}$ converges uniformly to f. ■

Proposition 2.68. *If X is complete, then $(F(S,X), d_\infty)$ is complete.*

Proof. Suppose that (X,d) is complete, and let $\{f_n\}_{n\in\mathbb{N}}$ be a Cauchy sequence in $(F(S,X), d_\infty)$. Take $0 < \varepsilon < 1$. Then, there is some N such that, for all $m,n > N$, $d_\infty(f_m, f_n) < \varepsilon$. Since $\varepsilon < 1$, we have that $\sup_{s\in S}\{d(f_m(s), f_n(s))\} < \varepsilon$, so that $d(f_m(s), f_n(s)) < \varepsilon$ for all $m,n > N$. Thus, for each $s \in S$, $\{f_n(s)\}_{n\in\mathbb{N}}$ is Cauchy in X, so it has a limit. Define $f(s) = \lim_{n\to\infty} f_n(s)$. We wish to show that $\{f_n\}_{n\in\mathbb{N}}$ converges to f in $F(S,X)$. Let $0 < \varepsilon < 1$, and take N such that $d_\infty(f_n, f_m) < \varepsilon$ for all $n,m > N$. Then, $d(f_n(s), f_m(s)) < \varepsilon$ for all $s \in S$ and $n,m > N$. Letting m tend to ∞, we have that $d(f_n(s), f(s)) < \varepsilon$ for all $s \in S$ and $n,m > N$, meaning that $d_\infty(f_n, f) < \varepsilon$ for all $n > N$. ■

Proposition 2.69. *If $\{f_n\}_{n\in\mathbb{N}}$ is a sequence of continuous functions converging uniformly to f, then f is continuous.*

Proof. Let $\varepsilon > 0$. Take $N \in \mathbb{N}$ such that $d(f_n(x), f(x)) < \varepsilon/3$ for all $n > N$ and $x \in X$. Fix $x \in X$. Since f_n is continuous, there is an open set $U \ni x$ such that $d(f_n(x), f_n(y)) < \varepsilon/3$ for all $y \in U$. Thus, for all $y \in U$ and $n > N$, we have

$$d(f(x), f(y)) \leq d(f(x), f_n(x)) + d(f_n(x), f_n(y)) + d(f_n(y), f(y)) < 3\varepsilon/3 = \varepsilon.$$

Thus, f is continuous at x, and since it was arbitrary, f is continuous. ■

Lemma 2.70. *Let X be a normal topological space, let Y be a closed subset of X, and let $f : Y \to \mathbb{R}$ be a continuous function such that $|f(x)| \leq a$ for all $x \in Y$. Then there exists a continuous function $g : X \to \mathbb{R}$ such that $|g(x)| \leq a/3$ for all $x \in X$, and $|f(y) - g(y)| < 2a/3$ for all $y \in Y$.*

Proof. Let $E_0 = \{y \in Y : f(y) \leq -a/3\}$ and $F_0 = \{y \in Y : f(y) \geq a/3\}$. Then E_0 and F_0 are disjoint closed subsets, so there is a continuous function $g : X \to \mathbb{R}$ such that $|g(x)| \leq a/3$ and $g = -a/3$ on E_0 and $g = a/3$ on F_0 (replace $[0,1]$ with $[-a/3, a/3]$ in the statement and proof of the theorem). For $y \in E_0$, $-a \leq f(y) \leq -a/3$, and $g(y) = -a/3$, so that $|f(y) - g(y)| \leq 2a/3$, and similarly for F_0. Now, take $y \in Y \setminus (E_0 \cup F_0)$. Then $|f(y)| < a/3$, and since $|g(y)| \leq a/3$, we have that $|f(y) - g(y)| \leq 2a/3$. ■

Theorem 2.71 (Tietze Extension Theorem)**.** *Let X be a normal topological space, and let Y be a closed subset of X. Let $f : Y \to \mathbb{R}$ be continuous and bounded. Then there exists a continuous and bounded $h : X \to \mathbb{R}$ such that $h = f$ on Y.*

Proof. We may assume that f is bounded by 1 without loss of generality. Then we have $g_0 : X \to \mathbb{R}$ with $|g_0(x)| \leq 1/3$ for all $x \in X$ and $|f(y) - g_0(y)| \leq 2/3$ for all $y \in Y$. Applying the lemma to $(f - g_0) : Y \to \mathbb{R}$, we get $g_1 : X \to \mathbb{R}$ with $|g_1(x)| \leq (1/3)(2/3)$ for all $x \in X$ and $|f(y) - g_0(y) - g_1(y)| \leq (2/3)^2$. Continuing, we get a sequence of functions $g_0, g_1, \ldots$ such that $|g_n(x)| \leq 1/3(2/3)^n$ for all $x \in X$ and $|f(y) - g_0(y) - \cdots - g_n(y)| \leq (2/3)^{n+1}$ for all $y \in Y$. Define $h_i = \sum_{n=0}^{i} g_n$. Then since $|g_n(x)| \leq (1/3)(2/3)^n$ and the sum $\sum_{n=0}^{\infty}(1/3)(2/3)^n$ converges, it follows that $\sum_{n=0}^{\infty} g_n$ converges (absolutely and uniformly). Thus, $\{h_i\}_{0 \leq i}$ converges uniformly to $h = \sum_{n=0}^{\infty} g_n$, which is continuous. Now,

$$|h| = \left|\sum_{n=0}^{\infty} g_n\right| \leq \sum_{n=0}^{\infty} |g_n| \leq \sum_{n=0}^{\infty} \frac{1}{3} \cdot \left(\frac{2}{3}\right)^n = 1.$$

Now, for $y \in Y$, we have that

$$|f(y) - h_i(y)| = \left|f(y) - \sum_{n=0}^{i} g_n(y)\right| \leq \left(\frac{2}{3}\right)^{i+1}.$$

Thus, as i tends to ∞, $|f(y) - h(y)| = 0$ for all $y \in Y$, so $h = f$ on Y. ■

2.7 Compactness

Definition 2.72. Let X be a topological space. A space $Y \subseteq X$ is compact if, for every open cover $\{U_i\}_{i \in I}$ of Y, there is a finite set $\{i_1, \ldots, i_n\}$ such that $Y \subseteq U_{i_1} \cup \cdots \cup U_{i_n}$.

Proposition 2.73. *If X is compact and $V \subseteq X$ is closed, then V is compact.*

Proof. Take an open cover $\{U_i\}_{i \in I}$ of V, and consider the open cover $\{U_i\}_{i \in I} \cup \{X \setminus V\}$ of X. This has a finite subcover $\{X \setminus V\} \cup U_{i_1} \cup \cdots \cup U_{i_n}$. Then, $V \subseteq U_{i_1} \cup \cdots \cup U_{i_n}$, so V is compact. ■

Proposition 2.74. *If $Y_1, \ldots, Y_n \subseteq X$ are compact, then $Y = Y_1 \cup \cdots \cup Y_n$ is compact.*

Proof. Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of Y. Define

$$A_i = \{\alpha \in A : U_\alpha \cap Y_i \neq \varnothing\}.$$

Then $A \supseteq A_1 \cup \cdots \cup A_n$, and each A_i is nonempty. Further, $\{U_\alpha\}_{\alpha \in A_i}$ is an open cover for Y_i. Take $B_i \subseteq A_i$ to be finite such that $\{U_\beta\}_{\beta \in B_i}$ is an open cover for Y_i. Let $B = B_1 \cup \cdots \cup B_n \subseteq A$. Then B is finite, and

$$Y = Y_1 \cup \cdots \cup Y_n \subseteq \bigcup_{\beta \in B_1} U_\beta \cup \cdots \cup \bigcup_{\beta \in B_n} U_\beta = \bigcup_{\beta \in B} U_\beta.$$

Thus, Y is compact. ■

Proposition 2.75. *If X is Hausdorff, and if $K \subseteq X$ is compact, then K is closed.*

Proof. Let $K \subseteq X$ be compact. We show that $X \setminus K$ is open. Let $x \in X \setminus K$. Since X is Hausdorff, for each $k \in K$, there is $U_k \ni k$ and $V_k \ni x$ with $U_k \cap V_k = \varnothing$, where U_k and V_k are open. Now, $K \subseteq \bigcup_{k \in K} U_k$. Since K is compact, there are finitely many $k_1, \ldots, k_n \in K$ such that $K \subseteq U_{k_1} \cup \cdots \cup U_{k_n} = U$. Define $V = V_{k_1} \cap \cdots \cap V_{k_n}$. Then $x \in V$, and $V \cap U = \varnothing$, so $V \cap K = \varnothing$. Hence, $V \subseteq X \setminus K$. Thus, for each $x \in X \setminus K$, there is an open neighborhood of x contained entirely in $X \setminus K$, so that $X \setminus K$ is open. Thus, K is closed. ■

Corollary 2.76. *A compact Hausdorff space is T_3.*

Proof. Let $x \in X$ and $K \subseteq X$ be a closed set not containing X. Since closed subsets of compact spaces are compact, K is compact. Continue as in the previous proof to obtain U and V. Then, $K \subseteq U$, $x \in V$, and $V \cap U = \varnothing$, so X is T_3. ■

Proposition 2.77. *A compact Hausdorff space is T_4.*

Proof. Let V_1 and V_2 be disjoint closed sets. Since X is compact, V_1 and V_2 are compact. Since X is T_3, for each $v \in V_2$, let U_v and U'_v be disjoint open sets with $V_1 \subseteq U_v$ and $v \in U'_v$. Then $V_2 \subseteq \bigcup_{v \in V_2} U'_v$, and since V_2 is compact, there are $v_1, \ldots, v_n$ such that $V_2 \subseteq U'_{v_1} \cup \cdots \cup U'_{v_n} = U'$. Define $U = U_{v_1} \cap \cdots \cap U_{v_n} \supseteq V_1$. Then $V_1 \subseteq U$, $V_2 \subseteq U'$, and $U \cap U' = \varnothing$, so X is T_4. ■

Proposition 2.78. *If $f : X \to Y$ is a continuous map from a compact space X to a topological space Y, then $f(X) \subseteq y$ is compact.*

Proof. Let $\{U_i\}_{i \in I}$ be an open cover of $f(X)$. Since f is continuous, $\{f^{-1}(U_i)\}_{i \in I}$ is an open cover of X. Thus, there is a finite set $\{i_1, \ldots, i_n\}$ such that $\{f^{-1}(U_{i_k})\}_{1 \leq k \leq n}$ is a subcover of X. Then, $\{U_{i_k}\}_{1 \leq k \leq n}$ is a finite subcover of $f(X)$. ■

Theorem 2.79. *Let f be a continuous map from a compact space X to a Hausdorff space Y. If f is injective, then f is a homeomorphism of X and $f(X)$.*

Proof. Firstly, $f : X \to f(X)$ is continuous and surjective. If f is injective, then it is bijective from X to $f(X)$. Thus, it suffices to show that f is open. Let $U \subseteq X$ be an open set. Then $X \setminus U$ is closed, and therefore compact. Thus, $f(X \setminus U) = f(X) \setminus f(U)$ is compact in $f(X)$. As $f(X)$ is Hausdorff, $f(X) \setminus f(U)$ is closed, so $f(U)$ is open. ■

2.8 Locally Compact Spaces

Definition 2.80. A topological space X is locally compact if, for each $p \in X$, there is an open set $U \ni p$ such that $\overline{U}$ is compact.

Example 2.81. $\mathbb{R}^n$ is locally compact, but not compact. Compact spaces are also locally compact.

Theorem 2.82 (One-Point Compactification)**.** *Let $(X, \mathscr{T}_X)$ be a locally compact Hausdorff topological space, and let X^+ be a set consisting of X and a point $\{\infty\}$ ($X^+ = X \cup \{\infty\}$). Then there exists a unique topology $\mathscr{T}_{X^+}$ for X^+ such that X^+ is a compact Hausdorff space, and the relative topology on $X \subseteq X^+$ coincides with $\mathscr{T}_X$.*

Proof. Define a collection of subsets $\mathscr{T}_{X^+}$ of X^+ by declaring $U \in \mathscr{T}_{X^+}$ if either

(1) $U \in \mathscr{T}_X$

(2) $\{\infty\} \in U$, and $X \setminus U$ is a compact subset of X.

We first check that the relative topology of $X \subseteq X^+$ coincides with $\mathscr{T}_X$. It suffices to check that subsets of $\mathscr{T}_{X^+}$ of type (2) intersect X in an open set, as this is automatically true for subsets of type (1). If $X \setminus U$ is compact in X, then it is closed in X, so $U \cap X$ is open in X. Now we verify that $\mathscr{T}_{X^+}$ is a topology for X^+. Clearly, $\varnothing \in \mathscr{T}_{X^+}$ and, since $X \setminus X^+ = \varnothing$ is compact, $X^+ \in \mathscr{T}_{X^+}$. Let $\{U_\alpha\}_{\alpha \in A}$ be a collection of sets in $\mathscr{T}_{X^+}$ and let $U = \bigcup_{\alpha \in A} U_\alpha$. If $U_\alpha \in \mathscr{T}_X$ for all α, then $U \in \mathscr{T}_X$ and so $U \in \mathscr{T}_{X^+}$. Otherwise, let $B \subseteq A$ be the set of indices for which U_β for $\beta \in B$ is type (2). Then

$$\begin{aligned} X \setminus U &= X \setminus \left(\bigcup_{\beta \in B} U_\beta \cup \bigcup_{\alpha \in A \setminus B} U_\alpha \right) = X \cap \left(\bigcup_{\beta \in B} U_\beta \cup \bigcup_{\alpha \in A \setminus B} U_\alpha \right)^c \\ &= X \cap \left(\bigcap_{\beta \in B} U_\beta^c \cap \bigcap_{\alpha \in A \setminus B} U_\alpha^c \right) = \bigcap_{\beta \in B} (X \setminus U_\beta) \cap \bigcap_{\alpha \in A \setminus B} (X \setminus U_\alpha). \end{aligned}$$

By assumption, all sets in the intersection on the left are compact (and hence closed), and the sets in the intersection on the right are closed, so $X \setminus U$ is a closed subset of $X \setminus U_{\beta_0}$ for some $\beta_0 \in B$, which is compact. Hence, $X \setminus U$ is compact, meaning $U \in \mathscr{T}_{X^+}$. Now let $U_1, \ldots, U_n \in \mathscr{T}_{X^+}$, and let $U = U_1 \cap \cdots \cap U_n$. If any of the U_i's are type (1), then $U \in \mathscr{T}_X$, and so $U \in \mathscr{T}_{X^+}$. Suppose that $U_1, \ldots, U_n$ are type (2). Then $\{\infty\} \in U$, and

$$\begin{aligned} X \setminus U &= X \setminus (U_1 \cap \cdots \cap U_n) = X \cap (U_1 \cap \cdots \cap U_n)^c \\ &= X \cap (U_1^c \cup \cdots \cup U_n^c) = X \setminus U_1 \cup \cdots \cup X \setminus U_n. \end{aligned}$$

Since a finite union of compact sets is compact, $X \setminus U$ is compact, so $U \in \mathscr{T}_{X^+}$. Hence, $\mathscr{T}_{X^+}$ is a topology. We now show that X^+ is compact. Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of X^+.

Then for some $\alpha_0 \in A$, we have that $\{\infty\} \in U_{\alpha_0}$. Now $\{U_\alpha \cap X\}_{\alpha \in A \setminus \{\alpha_0\}}$ is an open cover of the compact set $X \setminus U_{\alpha_0}$, so there is a finite subcover

$$X \setminus U_{\alpha_0} \subseteq U_{\alpha_1} \cup \cdots \cup U_{\alpha_n}.$$

Since $\{\infty\} \in U_{\alpha_0}$, we have

$$X^+ \subseteq U_{\alpha_0} \cup U_{\alpha_1} \cup \cdots \cup U_{\alpha_n}.$$

Thus X^+ is compact. Now we verify that X^+ is Hausdorff. Since X is Hausdorff, we only need to show that any $x \in X$ can be separated from $\{\infty\}$ by open sets. As X is locally compact, there is an open subset $U \ni x$ such that $\overline{U}$ is compact. Then, $X^+ \setminus \overline{U}$ is open, contains $\{\infty\}$, and is disjoint from U. Hence, X^+ is Hausdorff. Finally, we show that $\mathscr{T}_{X^+}$ is unique. Let $\mathscr{T}$ be another topology satisfying our assumptions. Let $U \in \mathscr{T}_{X^+}$ be of type (1). Since the relative topology of X in $(X^+, \mathscr{T})$ coincides with $\mathscr{T}_X$, there is an open set $V \in \mathscr{T}$ such that $X \cap V = U$. Note that X is $\mathscr{T}$-open, as $\{\infty\} = X^+ \setminus X$ must be compact (it is finite), and hence closed. Since X and V are both $\mathscr{T}$-open, $U = X \cap V$ is $\mathscr{T}$-open. If U is of type (2), then $\{\infty\} \in U$ and $X \setminus U$ is compact in $(X, \mathscr{T}_X)$ and thus also in $(X^+, \mathscr{T})$. Hence $X \setminus U$ is $\mathscr{T}$-closed, so U is $\mathscr{T}$-open. Thus, $\mathscr{T}_{X^+} \subseteq \mathscr{T}$. This shows that $\mathrm{id} : (X^+, \mathscr{T}) \to (X^+, \mathscr{T}_{X^+})$ is continuous. Hence, id is a continuous bijection from a compact space to a Hausdorff space, and so it is a homeomorphism. Thus, $\mathscr{T} = \mathscr{T}_{X^+}$. ∎

2.9 Connectedness and Path Connectedness

Definition 2.83. A set is clopen if it is both closed and open.

Definition 2.84. A topological space X is connected if the only clopen sets are $\varnothing$ and X. If there does exist a clopen set $U \neq \varnothing, X$, X is disconnected.

Proposition 2.85. *A space X is disconnected if and only if there exist disjoint open sets $U, V \neq \varnothing, X$ such that $X = U \cup V$.*

Proof. Suppose that X is disconnected. Take $U \neq \varnothing, X$ to be clopen, and take $V = X \setminus U$. Now suppose that there exist such open sets U and V. Then $U = X \setminus V$, so U is closed, and hence clopen. ∎

Proposition 2.86. *If X is connected and $f : X \to Y$ is continuous, then $f(X)$ is connected.*

Proof. Suppose that $f(X)$ was disconnected. Take disjoint open sets $E, F \neq \varnothing, f(X)$ such that $f(X) = E \cup F$. Then $X = f^{-1}(E) \cup f^{-1}(F)$ and $f^{-1}(E)$, $f^{-1}(F)$ are disjoint open sets, a contradiction. ∎

Theorem 2.87. *Any interval in $\mathbb{R}$ is connected.*

Proof. Since any interval is homeomorphic to $[0,1]$, $[0,1)$ or $(0,1)$, we only need to prove these are connected. Suppose that $[0,1]$ was not connected, so that there are nonempty disjoint open sets E and F such that $[0,1] = E \cup F$. Without loss of generality, assume $1 \notin E$. Let $t_0 = \sup_{t \in E}\{t\}$. Since E is closed, $t_0 \in E$. Also, since $1 \notin E$, $t_0 < 1$. But since E is open, there is $\varepsilon > 0$ such that $[t_0, t_0 + \varepsilon) \subseteq E$, a contradiction. Replacing 1 with 0, $[0,1)$ is connected. Working in $(0,1)$, t_0 is guaranteed to be less than 1, so we are done. ∎

Definition 2.88. Let X be a topological space. A path (from x_0 to x_1) is a continuous function $\gamma : [0,1] \to X$ (with $\gamma(0) = x_0$ and $\gamma(1) = x_1$).

Remark 2.89. The image of a path is connected.

Definition 2.90. A space X is path connected if, for all $x_0, x_1 \in X$, there exists a path γ from x_0 to x_1.

Proposition 2.91. *The relation "there is a path from x to y" is an equivalence relation on X.*

Proof. The constant map $\alpha : [0,1] \to X$ given by $\alpha(s) = x$ is a path from x to x. Suppose that γ is a path from x to y. Then $\beta(s) = \gamma(1-s)$ is a path from y to x. Now suppose that α is a path from x to y, and that β is a path from y to z. Then

$$\gamma(s) = \begin{cases} \alpha(2s) & 0 \leq s \leq \frac{1}{2} \\ \beta(2s-1) & \frac{1}{2} \leq s \leq 1 \end{cases}$$

is a path from x to z. ∎

Definition 2.92. The path components of X are the equivalence classes of the relation above.

Remark 2.93. A space is path connected if and only if there is exactly one path component.

Proposition 2.94. *Let $\{E_\alpha\}_{\alpha \in A}$ be a collection of connected subsets of a topological space X such that, for all $\alpha \neq \beta$, $E_\alpha \cap E_\beta \neq \varnothing$. Then $\bigcup_{\alpha \in A} E_\alpha$ is connected.*

Proof. Let $E = \bigcup_{\alpha \in A} E_\alpha$, and suppose that we have disjoint open subsets U and V such that $E = U \cup V$. We show either $U = E$ or $V = E$. For each $\alpha \in A$, we may write $E_\alpha = (E_\alpha \cap U) \cup (E_\alpha \cap V)$. This is a disjoint open cover of E_α. Since E_α is connected, either $E_\alpha = E_\alpha \cap U$ or $E_\alpha = E_\alpha \cap V$. In other words, $E_\alpha \subseteq U$ or $E_\alpha \subseteq V$. Suppose that $E_\alpha \subseteq U$. Then, for $\beta \neq \alpha$, we have $E_\beta \subseteq U$ or $E_\beta \subseteq V$. If $E_\beta \subseteq V$, then $\varnothing \neq E_\alpha \cap E_\beta \subseteq U \cap V$, so U and V are not disjoint. Thus, $E_\beta \subseteq U$. Hence, $E = \bigcup_{\alpha \in A} E_\alpha \subseteq U$. Since $U \subseteq E$, $E = U$. ∎

Corollary 2.95. *The path components of a topological space are connected.*

Proof. Denote the path component by P, and fix $x \in P$. For each $y \in P$, take γ_y to be the path from x to y. Write Γ_y for the image of γ_y. Then

$$P \subseteq \bigcup_{y \in P} \Gamma_y.$$

Now suppose that $z \in \Gamma_y$, say $z = \gamma_y(t_0)$. Then $z \in P$, as $\gamma_z(t) = \gamma_y(t_0 \cdot t)$. Hence, $\Gamma_y \subseteq P$ for all y, so

$$P = \bigcup_{y \in P} \Gamma_y.$$

Now, each Γ_y is connected, and they all intersect at x, so P is connected. ∎

Corollary 2.96. *If a space is path connected, it is connected.*

Definition 2.97. The connected component of $x \in X$ is the union of all connected subsets of X containing x.

Proposition 2.98. *Connected components are either equal or disjoint.*

Proof. Let $x, y \in X$, and let E_x and E_y denote their connected components. If $E_x \cap E_y \neq \varnothing$, then $E_x \cup E_y$ is a connected subset of X containing x and y. Thus, $E_x \cup E_y \subseteq E_x$, so $E_y \subseteq E_x$. Similarly, $E_x \subseteq E_y$, so $E_x = E_y$. ■

2.10 Zorn's Lemma

Definition 2.99. A partially ordered set (a poset) is a nonempty set P with a relation $\leq$ such that

(1) For all $x \in P$, $x \leq x$.

(2) If $x \leq y$ and $y \leq z$, then $x \leq z$.

(3) If $x \leq y$ and $y \leq x$, then $x = y$.

Definition 2.100. A totally ordered set is a post P such that, for all $x, y \in P$, either $x \leq y$ or $y \leq x$.

Definition 2.101. An upper bound for a subset $S \subseteq P$ is an element $x \in P$ such that $y \leq x$ for all $y \in S$. An element $m \in P$ is a maximal element if, for all $y \neq m \in P$, $m \not\leq y$.

Theorem 2.102 (Zorn's Lemma)**.** *If P is a poset such that every totally ordered subset has an upper bound, then P has a maximal element.*

Example 2.103. We show that every vector space has a basis using Zorn's Lemma. Let P be the collection of linearly independent subsets of a vector space V. Then this collection is partially ordered by set inclusion $\subseteq$. Let $\{L_i\}_{i \in I}$ be a totally ordered subset of P, and let $L = \bigcup_{i \in I} L_i$. This is an upper bound for $\{L_i\}_{i \in I}$, but we must show that $L \in P$ to apply Zorn's Lemma. If $\{v_1, \ldots, v_n\} \subseteq L$, then for each $1 \leq j \leq n$, there is an i_j such that $v_j \in L_{i_j}$. Since $\{L_i\}_{i \in I}$ is totally ordered, there is m such that $L_{i_1}, L_{i_2}, \ldots, L_{i_n} \subseteq L_m$. Then $\{v_1, \ldots, v_n\} \subseteq L_m$, which is linearly independent. Thus $L \in P$. By Zorn's Lemma, there is a maximal element $B \in P$. We claim that B is a basis. Suppose that it did not. Then $\mathrm{span}(B) \subseteq V$ must be a proper subspace. Thus, there is $v \in V \setminus \mathrm{span}(B)$. Then $B \cup \{v\}$ is linearly independent and contains B, so B is not maximal.

2.11 Infinite Product Spaces

Definition 2.104. A family S of open subsets of a topological space X is a subbase for the topology of X if the collection of finite intersections of sets in S forms a basis for the topology.

Example 2.105. In $\mathbb{R}$, $\{(a,b) : a < b \in \mathbb{R}\}$ is a basis for the topology. We also have a subbase $\{(-\infty, a) : a \in \mathbb{R}\} \cup \{(a,\infty) : a \in \mathbb{R}\}$.

Lemma 2.106 (Alexander Subbase Theorem)**.** *Let S be a subbase for X. If every open cover of X by elements of S has a finite subcover, then X is compact.*

Proof. Suppose that X is not compact. We show that there exists an open cover from S with no finite subcover. Take an open cover C of X with no finite subcover. Let P be the collection of open covers $\mathcal{E}$ such that no finite subcover exists. This is nonempty, and partially ordered by inclusion. Let $\{\mathcal{E}_\alpha\}_{\alpha\in A}$ be a totally ordered subset of P, and set $\mathcal{E} = \bigcup_{\alpha\in A} \mathcal{E}_\alpha$. Since each $\mathcal{E}_\alpha$ is a cover, $\mathcal{E}$ is a cover. If $\{U_1, \ldots, U_n\} \subseteq \mathcal{E}$, then since $\{\mathcal{E}_\alpha\}_{\alpha\in A}$ is totally ordered, there is α such that $\{U_1, \ldots, U_n\} \subseteq \mathcal{E}_\alpha$ (for the same reason as in the proof that every vector space has a basis). Thus, $\{U_1, \ldots, U_n\}$ cannot be a cover of X. Hence, $\mathcal{E} \in P$, so we may apply Zorn's Lemma. By Zorn's Lemma, P has a maximal element M. Take $M_0 = M \cap S$. If M_0 is a cover of X, then it is an open cover of X by elements of S with no finite subcover, so it suffices to show that M_0 covers X. Let $x \in X$. Since M covers X, there is $U \in M$ such that $x \in U$. Since S is a subbase, there are $V_1, \ldots, V_n \in S$ such that $x \in V_1 \cap \cdots \cap V_n \subseteq U$. Suppose that $V_i \notin M$ for each i. Then, $\{V_i\} \cup M$ is a cover of X which must have a finite subcover, as M is maximal. Then there are $W_{i,1}, \ldots, W_{i,m_i} \in M$ such that $\{V_i, W_{i,1}, \ldots, W_{i,m_i}\}$ covers X. Repeating this for each i, we have that

$$\{V_1 \cap \cdots \cap V_n, W_{1,1}, \ldots, W_{1,m_1}, \ldots, W_{n,1}, \ldots, W_{n,m_n}\} \subseteq M$$

is a finite subcover of X, a contradiction. Thus, there is $V_i \in M$. Then, $V_i \in M_0$. Since $x \in V_1 \cap \cdots \cap V_n$, we have that $x \in V_i$, so M_0 covers X. ■

Definition 2.107. Let A be an index set such that, for each $\alpha \in A$, X_α is a nonempty set. The Cartesian Product

$$X = \prod_{\alpha\in A} X_\alpha$$

is the set of all functions $x : A \to \bigcup_{\alpha\in A} X_\alpha$ with $x(\alpha) \in X_\alpha$ for all $\alpha \in A$.

Remark 2.108. For each $\alpha \in A$, there is a natural projection

$$\pi_\alpha : X \to X_\alpha$$

by $\pi_\alpha(x) = x(\alpha)$.

Definition 2.109. Let $\{X_\alpha\}_{\alpha\in A}$ be a set of topological spaces. The product topology on $X = \prod_{\alpha\in A} X_\alpha$ is the topology given by the basis

$$\mathcal{B} = \{\{x \in X : x(\alpha_j) \in U_{\alpha_j}, 1 \le j \le m\} : U_{\alpha_j} \subseteq X_{\alpha_j} \text{ is open}\}.$$

Remark 2.110. If $U = \{x \in X : x(\alpha_j) \in U_{\alpha_j}\}$, then $U = \pi_{\alpha_1}^{-1}(U_{\alpha_1}) \cap \cdots \cap \pi_{\alpha_m}^{-1}(U_{\alpha_m})$, so $\{\pi_\alpha^{-1}(U_\alpha) : \alpha \in A, U_\alpha \subseteq X_\alpha \text{ is open}\}$ is a subbase for the product topology.

Theorem 2.111 (Tychonoff)**.** *Any product of compact spaces is compact.*

Proof. Let $X = \prod_{\alpha \in A} X_\alpha$ with each X_α compact. Let D be a family of subbasic sets of the form $\pi_\alpha^{-1}(U_\alpha)$, where $\alpha \in A$ and $U_\alpha \subseteq X_\alpha$ is open. We show that if no finite collection of sets in D covers X, then D does not cover X (this means that if D covers X, it must have a finite subcover, and since D is arbitrary, this suffices by the Alexander Subbase Theorem). Suppose that no finite collection in D covers X. Fix an index $\beta \in A$ and consider the elements of D of the form $\pi_\beta^{-1}(V_i)$ where $V_i \subseteq X_\beta$ are open. If the collection of these V_i's cover X_β, then since X_β is compact, we would have a finite collection $\{V_{i_1}, \ldots, V_{i_n}\}$ which cover X_β, hence $\{\pi_\beta^{-1}(V_{i_1}), \ldots, \pi_\beta^{-1}(V_{i_n})\}$ would be a finite subcover of X from D, a contradiction. Thus, the V_i's do not cover X_β, and hence, we can choose $x_\beta \in X_\beta$ such that $x_\beta \in X \setminus \bigcup_i V_i$. Now define an element of $\prod_{\alpha \in A} X_\alpha$ by $f : A \to \bigcup_{\alpha \in A} X_\alpha$ by $f(\beta) = x_\beta$ for each $\beta \in A$, where x_β is as described above. By construction, this is not contained in any set of D. ■

3 Homotopy Theory

3.1 Groups

Definition 3.1. A group is a set G with an operation $\cdot : G \times G \to G$, where $\cdot(g, h) = g \cdot h$, such that

- $g \cdot (h \cdot k) = (g\dot{h}) \cdot k$ for all $g, h, k \in G$.
- There exists an element $e \in G$ such that $g \cdot e = e \cdot g = g$ for all $g \in G$.
- For all $g \in G$, there exists $g' \in G$ such that $g \cdot g' = g' \cdot g = e$.

Proof. The identity element is unique, and inverses are unique. ■

Proof. Suppose that e_1, e_2 are two identity elements. Then $e_1 = e_1 e_2 = e_2$. Suppose that h and k are two inverses of g. Then $h = h(gk) = (hg)k = k$. ■

Definition 3.2. A nonempty subset $H \subseteq G$ is a subgroup if it is closed under multiplication and closed under inverses.

Definition 3.3. Let G and H be groups. A map $f : G \to H$ is a (group) homomorphism if $f(g_1 \cdot g_2) = f(g_1) \cdot f(g_2)$ for all $g_1, g_2 \in G$.

Proposition 3.4. *If $f : G \to H$ is a homomorphism, then*

(1) $f(e_G) = e_H$.

(2) $f(g^{-1}) = f(g)^{-1}$.

Proof.

(1) We have that $f(e_G) = f(e_G \cdot e_G) = f(e_G) \cdot f(e_G)$. Multiplying by $f(e_G)^{-1}$, $e_H = f(e_G)$.

(2) We have that $f(g) \cdot f(g^{-1}) = f(g \cdot g^{-1}) = f(e_G) = e_H$ and that $f(g^{-1}) \cdot f(g) = f(g^{-1} \cdot g) = f(e_G) = e_H$. Since inverses are unique, $f(g)^{-1} = f(g^{-1})$.

■

3.2 Homotopic Paths

Definition 3.5. Let X and Y be topological spaces, and let $f, g : X \to Y$ be maps. A homotopy from f to g is a continuous map $F : X \times [0, 1] \to Y$ such that $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$. We say the maps f and g are homotopic and write $f \simeq g$.

Notation. Let $f, g : X \to Y$ and let F be a homotopy from f to g. We may write $f_t(x) = F(x, t)$.

Lemma 3.6 (Glung Lemma). *Let $f : X \to Y$ be a map between two topological spaces. Let $E, F \subseteq X$ be two closed (or open) subsets such that $X = E \cup F$. If the restrictions $f|E$ and $f|F$ are continuous, then f is continuous.*

Proof. Let $V \subseteq Y$ be closed (respectively open). Then $f|_E^{-1}(V) = f^{-1}(V) \cap E$ is closed (respectively open). Similarly, $f|F^{-1}(V)$ is closed (respectively open). Hence,

$$f^{-1}(V) = f^{-1}(V) \cap (E \cup F) = (f^{-1}(V) \cap E) \cup (f^{-1}(V) \cap F) = f|_E^{-1}(V) \cup f|F^{-1}(V),$$

which is a finite union of closed (respectively open) sets, and is thus closed (respectively open). Hence, f is continuous. ∎

Proposition 3.7. *Being homotopic is an equivalence relation on the set of maps from X to Y.*

Proof. Clearly, $f \simeq f$: define $F(x, t) = f(x)$. If $f \simeq g$, say where F is a homotopy from f to g, define $G(x, t) = F(x, 1 - t)$. Finally, if $f \simeq g$ and $g \simeq h$, where F is a homotopy from f to g and G is a homotopy from g to h. Define

$$H(x,t) = \begin{cases} F(x, 2t) & 0 \le t \le \frac{1}{2} \\ G(x, 2t-1) & \frac{1}{2} \le t \le 1 \end{cases}.$$

This is a homotopy from f to h. ∎

Definition 3.8. If $A \subseteq X$ is closed and $f, g : X \to Y$ are such that $f(a) = g(a)$ for all $a \in A$, then we say f is homotopic to g relative to A if there exists a homotopy $F : X \times [0, 1] \to Y$ such that, for all $a \in A$, $F(a, t) = f(a) = g(a)$.

Proposition 3.9. *Being homotopic relative to A is an equivalence relation.*

Proof. Define $F(x, t) = f(x)$ to show reflexivity. If $f \simeq g$, say where F is a homotopy from f to g, define $G(x, t) = F(x, 1 - t)$. Finally, if $f \simeq g$ and $g \simeq h$, where F is a homotopy from f to g and G is a homotopy from g to h. Define

$$H(x,t) = \begin{cases} F(x, 2t) & 0 \le t \le \frac{1}{2} \\ G(x, 2t-1) & \frac{1}{2} \le t \le 1 \end{cases}.$$

This is a homotopy from f to h. ∎

Definition 3.10. A path in X based at x is a continuous map $\gamma : [0, 1] \to X$ with $\gamma(0) = x$.

Definition 3.11. Let α be a path from x to y. The set of paths from x to y that are homotopic to α (with endpoints fixed: $F(0,t) = \alpha(0)$ and $F(1,t) = \alpha(1)$ for all t) is called the homotopy class of α, and is denoted $[\alpha]$. If $[\alpha] = [\beta]$, we may write $\alpha \simeq_{\{0,1\}} \beta$.

Proposition 3.12. *Let γ be a path in X from a to b, and let $p : [0,1] \to [0,1]$ be continuous such that $p(0) = 0$ and $p(1) = 1$. Then $[\gamma \circ p] = [\gamma]$.*

Proof. Consider the "straight line" homotopy from $\mathrm{id} : [0,1] \to [0,1]$ to $p : [0,1] \to [0,1]$ given by

$$\alpha_t(s) = (1-t)s + tp(s).$$

The composition of this homotopy with γ gives

$$\gamma(\alpha_t(s)) = \gamma((1-t)s + tp(s)).$$

At $t = 0$, this is γ, and at $t = 1$, this is $\gamma \circ p$. Hence, this is a homotopy from γ to $\gamma \circ p$. ∎

Definition 3.13. If α and β are paths in X with $\alpha(1) = \beta(0)$, then the concatenation of α and β is the path

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t) & 0 \le t \le \frac{1}{2} \\ \beta(2t-1) & \frac{1}{2} \le t \le 1 \end{cases}.$$

Proposition 3.14. *If $[\alpha_0] = [\alpha_1]$ and $[\beta_0] = [\beta_1]$, then, if it is defined,*

$$[\alpha_0 * \beta_0] = [\alpha_1 * \beta_1].$$

Proof. If $\{\alpha_t\}_{0 \le t \le 1}$ is a homotopy from α_0 to α_1 and $\{\beta_t\}_{0 \le t \le 1}$ is a homotopy from β_0 to β_1, then $\{\alpha_t * \beta_t\}_{0 \le t \le 1}$ is a homotopy from $\alpha_0 * \beta_0$ to $\alpha_1 * \beta_1$. ∎

Proposition 3.15. *Let α, β, γ be paths in X so that $(\alpha * \beta) * \gamma$ and $\alpha * (\beta * \gamma)$ are defined. Then,*

$$(\alpha * \beta) * \gamma \simeq_{\{0,1\}} \alpha * (\beta * \gamma).$$

Proof. We have that

$$\big((\alpha * \beta) * \gamma\big)(s) = \begin{cases} \alpha(4s) & 0 \le s \le \frac{1}{4} \\ \beta(4s-1) & \frac{1}{4} \le s \le \frac{1}{2} \\ \gamma(2s-1) & \frac{1}{2} \le s \le 1 \end{cases}$$

and that

$$\big(\alpha * (\beta * \gamma)\big)(s) = \begin{cases} \alpha(2s) & 0 \le s \le \frac{1}{2} \\ \beta(4s-2) & \frac{1}{2} \le s \le \frac{3}{4} \\ \gamma(4s-3) & \frac{3}{4} \le s \le 1 \end{cases}.$$

Consider $p : [0,1] \to [0,1]$ given by

$$p(s) = \begin{cases} \frac{s}{2} & 0 \le s \le \frac{1}{2} \\ s - \frac{1}{4} & \frac{1}{2} \le s \le \frac{3}{4} \\ 2s - 1 & \frac{3}{4} \le s \le 1 \end{cases}.$$

This is continuous, and

$$\Big(\big((\alpha*\beta)*\gamma\big)\circ p\Big)(s)=\begin{cases}\alpha(2s) & 0\le s\le \frac{1}{2}\\ \beta(4s-2) & \frac{1}{2}\le s\le\frac{3}{4}\\ \gamma(4s-3) & \frac{3}{4}\le s\le 1\end{cases}=\big(\alpha*(\beta*\gamma)\big)(s).$$

Hence, $\big((\alpha*\beta)*\gamma\big)$ and $\big(\alpha*(\beta*\gamma)\big)$ are homotopic. ■

Proposition 3.16. *If C_x is the constant path at x, then*

$$C_{\alpha(0)}*\alpha\simeq_{\{0,1\}}\alpha\simeq_{\{0,1\}}\alpha*C_{\alpha(1)}.$$

Proof. Note that

$$(C_{\alpha(0)}*\alpha)(t)=\begin{cases}\alpha(0) & 0\le t\le\frac{1}{2}\\ \alpha(2t-1) & \frac{1}{2}\le t\le 1\end{cases}.$$

Define $p:[0,1]\to[0,1]$ by $p(t)=\frac{t+1}{2}$. Then, $\big((C_{\alpha(0)}*\alpha)\circ p\big)(t)=\alpha(t)$, so $C_{\alpha(0)}*\alpha\simeq_{\{0,1\}}\alpha$. If $q(t)=\frac{t}{2}$, then similarly $\big((\alpha*C_{\alpha(1)})\circ q\big)(t)=\alpha(t)$. ■

Proposition 3.17. *If $\overline{\alpha}(t)=\alpha(1-t)$, then $\alpha*\overline{\alpha}\simeq_{\{0,1\}}C_{\alpha(0)}$ and $\overline{\alpha}*\alpha\simeq_{\{0,1\}}C_{\alpha(1)}$.*

Proof. Define

$$F(s,t)=\begin{cases}\alpha(2ts) & 0\le s\le\frac{1}{2}\\ \alpha(2t-2ts) & \frac{1}{2}\le s\le 1\end{cases}.$$

Then $F(s,0)=C_{\alpha(0)}(s)$ and $F(s,1)=\alpha*\overline{\alpha}(s)$. Hence, $\alpha*\overline{\alpha}\simeq_{\{0,1\}}C_{\alpha(0)}$. Now, notice that $\alpha(t)=\overline{\alpha}(1-t)$, so that $\overline{\alpha}*\alpha\simeq_{\{0,1\}}C_{\overline{\alpha}(0)}=C_{\alpha(1)}$. ■

3.3 The Fundamental Group

Definition 3.18. A loop in X based at x is a path $\gamma:[0,1]\to X$ such that $\gamma(0)=\gamma(1)=x$.

Definition 3.19. The fundamental group of X based at x is

$$\pi_1(X,x)=\big\{[\gamma]:\gamma\text{ is a loop based at }x\big\}$$

under the operation

$$[\alpha][\beta]=[\alpha*\beta].$$

Proposition 3.20. *$\pi_1(X,x)$ forms a group.*

Proof. We have that

- $\big([\alpha][\beta]\big)[\gamma]=[\alpha*\beta][\gamma]=[(\alpha*\beta)*\gamma]=[\alpha*(\beta*\gamma)]=[\alpha][\beta*\gamma]=[\alpha]\big([\beta][\gamma]\big)$.
- $[\alpha][C_x]=[\alpha*C_x]=[\alpha]=[C_x*\alpha]=[C_x][\alpha]$.
- $[\alpha][\overline{\alpha}]=[\alpha*\overline{\alpha}]=[C_x]=[\overline{\alpha}*\alpha]=[\overline{\alpha}][\alpha]$.

■

Notation. Let α and β be paths such that $\alpha(1) = \beta(0)$. We write $[\alpha][\beta] = [\alpha * \beta]$.

Definition 3.21. Let $x_0, x_1 \in X$, and let γ be a path from x_1 to x_o. We define

$$\begin{aligned} h_\gamma : \pi_1(X, x_0) &\to \pi_1(X, x_1) \\ [\alpha] &\mapsto [\gamma * \alpha * \overline{\gamma}]. \end{aligned}$$

Proposition 3.22. *The map h_γ is well defined.*

Proof. Suppose that $[\alpha] = [\alpha']$. Since $[\gamma] = [\gamma]$, $[\gamma * \alpha] = [\gamma * \alpha']$. Since $[\overline{\gamma}] = [\overline{\gamma}]$, $[\gamma * \alpha * \overline{\gamma}] = [\gamma * \alpha' * \overline{\gamma}]$. ■

Theorem 3.23. *Let $x_0, x_1 \in X$ and γ be a path in X from x_1 to x_0. Then the map $h_\gamma : \pi_1(X, x_0) \to \pi_1(X, x_1)$ is an isomorphism.*

Proof. We have that

$$\begin{aligned} h_\gamma([\beta][\eta]) = h_\gamma([\beta * \eta]) &= [\gamma * \beta * \eta * \overline{\gamma}] = [\gamma * \beta][\eta * \overline{\gamma}] \\ &= [\gamma * \beta][\overline{\gamma} * \gamma][\eta * \overline{\gamma}] = [\gamma * \beta * \overline{\gamma}][\gamma * \eta * \overline{\gamma}] = h_\gamma([\beta])h_\gamma([\eta]), \end{aligned}$$

so it remains to show that h_γ is bijective. Now suppose that $h_\gamma([\beta]) = h_\gamma([\eta])$. Then $[\gamma * \beta * \overline{\gamma}] = [\gamma * \eta * \overline{\gamma}]$. Since $[\gamma] = [\gamma]$ and $[\overline{\gamma}] = [\overline{\gamma}]$, we have that

$$[\beta] = [\overline{\gamma} * \gamma][\beta][\gamma * \overline{\gamma}] = [\overline{\gamma}][\gamma * \beta * \overline{\gamma}][\gamma] = [\overline{\gamma}][\gamma * \eta * \overline{\gamma}][\gamma] = [\overline{\gamma} * \gamma][\eta][\gamma * \overline{\gamma}] = [\eta].$$

Hence h_γ is injective. Let λ be a loop at x_1. Then $\overline{\gamma} * \gamma * \gamma$ is a loop at x_0, and

$$h_\gamma([\overline{\gamma} * \lambda * \gamma]) = [\gamma][\overline{\gamma} * \lambda * \gamma][\overline{\gamma}] = [\gamma * \overline{\gamma}][\lambda][\gamma * \overline{\gamma}] = [\lambda],$$

so h_γ is surjective. ■

Notation. Suppose that X is path connected. We write $\pi_1(X)$ for the group $\pi_1(X, x)$, where $x \in X$ is arbitrary.

Definition 3.24. Let X be a path connected topological space with $\pi_1(X) = \{1\}$. We say X is simply connected.

3.4 Induced Homomorphisms

Proposition 3.25. *Let X and Y be topological spaces, and let $f : X \to Y$ be a continuous map, and let $\alpha_0, \alpha_1 : [0, 1] \to X$ be paths such that $\alpha_0 \simeq_{\{0,1\}} \alpha_1$. Then $(f \circ \alpha_0) \simeq_{\{0,1\}} (f \circ \alpha_1)$.*

Proof. Compose a homotopy from α_0 to α_1 with f to obtain a homotopy from $f \circ \alpha_0$ to $f \circ \alpha_1$. ■

Definition 3.26. LEt X and Y be topological spaces, and let $f : X \to Y$ be continuous. Suppose that $f(x) = y$. We define $f_* : \pi_1(X, x) \to \pi_1(Y, y)$ by $f_*([\alpha]) = [f \circ \alpha]$, which is well-defined.

Theorem 3.27. *Let $f : X \to Y$ be a continuous map such that $f(x) = y$ for some fixed $x \in X$ and $y \in Y$. Then $f_* : \pi_1(X, x) \to \pi_1(Y, y)$ is a homomorphism. Moreover, if $X = Y$ and $x = y$, and f is the identity, then f_* is the identity isomorphism.*

Proof. If $\alpha * \beta$ is defined, then $f \circ (\alpha * \beta) = (f \circ \alpha) * (f \circ \beta)$. Hence, $f_*([\alpha][\beta]) = f_*([\alpha]) f_*([\beta])$. The second statement follows from definitions. ■

Proposition 3.28. *Let X, Y, W be topological spaces, and let $f : X \to Y$ and $g : Y \to W$ be continuous maps. Suppose further that $f(x) = y$ and $f(y) = w$ for fixed $x \in X$, $y \in Y$ and $w \in W$. Then $(g \circ f)_* = g_* \circ f_*$.*

Proof. We have that

$$(g \circ f)_*([\alpha]) = [g \circ f \circ \alpha] = g_*([f \circ \alpha]) = g_*(f_*([\alpha])) = (g_* \circ f_*)([\alpha]).$$

■

3.5 Covering Spaces

Definition 3.29. Let E and X be topological spaces, and $p : E \to X$ be continuous. An open subset $U \subseteq X$ is evenly covered by p if $p^{-1}(U)$ is a union of disjoint open subsets of E, each of which is mapped homeomorphically by p to U. The map p is a covering map if it is surjective, and if each $x \in X$ has an open neighborhood that is evenly covered by p. We say E is a covering space of X.

Definition 3.30. Let $p : E \to X$ be a covering map. Each $x \in X$ has an open neighborhood U such that $p^{-1}(U)$ is homeomorphic to $p^{-1}(x) \times U$. The subsets of $p^{-1}(U)$ which map homeomorphically to U are called sheets. If U is connected, the sheets of U are the connected components of $p^{-1}(U)$.

Notation. We will consider the circle, $\mathbb{S}^1$, to be a subset of the complex numbers.

Example 3.31. Let $p : \mathbb{R} \to \mathbb{S}^1$ be the map $p(t) = e^{2\pi i t}$. Then $p^{-1}(e^{2\pi i t_0}) = \{t_0 + k : k \in \mathbb{Z}\}$. If $U = \{e^{2\pi i t} : a < t < b\}$, then $p^{-1}(U) \bigcup_{k \in \mathbb{Z}} (a + k, b + k)$. If $b - a < 1$, then this a disjoint union, and each of these intervals is homeomorphically mapped to U by p. If $0 < \varepsilon < 1/2$, taking $U = \{e^{2\pi i t} : |t - t_0| < \varepsilon\}$, we see that p is a covering map.

Definition 3.32. Let $p : E \to X$ be a covering map, and $f : Y \to X$ be a continuous map. A continuous map $g : Y \to E$ such that $p \circ g = f$ is called a lift of f.

Lemma 3.33. *Let $p : E \to X$ be a covering map, and let Y be a connected topological space. Let $f : Y \to X$ be a continuous map, and let $g, h : Y \to E$ be lifts of f. If $g(y) = h(y)$ for some $y \in Y$, then $g = h$.*

Proof. Let $S = \{y \in Y : g(y) = h(y)\}$ and let $T = \{y \in Y : g(y) \neq h(y)\}$. Clearly, $Y = S \cup T$ and $S \cap T = \varnothing$. We show that if S is nonempty, then $S = Y$. Since, if $S = \varnothing$, $T = Y$, we show that either $S = Y$ or $T = Y$. We show that both S and T are open, and hence clopen, meaning that one must be Y, and the other must be empty. Take $y \in Y$, let $U \subseteq X$ be an evenly covered neighborhood of $f(y)$, and let V, W be sheets of $p^{-1}(U)$ with

$g(y) \in V$ and $h(y) \in W$. Note that if $g(y) = h(y)$, then $V = W$, and if $g(y) \neq h(y)$, then $V \cap W = \varnothing$. Since g and h are continuous, there is an open neighborhood $y \in N \subseteq Y$ such that $g(N) \subseteq V$ and $h(N) \subseteq W$ (take $N = g^{-1}(V) \cap h^{-1}(W)$, for instance). If $y \in T$, then $V \cap W = \varnothing$, so $g(z) \neq h(z)$ for all $z \in N$. Hence, we have found an open neighborhood of y contained in T, so T is open. If $y \in S$, then p maps the sheets homeomorphically (and hence bijectively) to U, meaning that, for $z \in N$, $g(z) \in V = W$ is the unique point in V such that $p(g(z)) = f(z)$, so that $g(z) = h(z)$ for $z \in N$. Hence, S is also open. ■

Theorem 3.34 (Path Lifting Theorem). *Let $p : E \to X$ be a covering map, and let $\gamma : [0,1] \to X$ be a path. Take $e_0 \in E$ such that $p(e_0) = \gamma(0)$. Then there exists a unique path $\alpha : [0,1] \to E$ such that $\alpha(0) = e_0$ and $p \circ \alpha = \gamma$.*

Proof. For each $x \in X$, let $U(x)$ be an evenly covered neighborhood. Then the preimages $\gamma^{-1}(U(x))$ form an open cover of $[0,1]$. Since $[0,1]$ is compact, we can find $0 = s_0 < s_1 < \cdots < s_m = 1$ and evenly covered open sets $U_1, \ldots, U_m$ such that $\gamma([s_{j-1}, s_j]) \subseteq U_j$ for each j. Since $p(e_0) = \gamma(0)$, there is an open neighborhood V_1 of e_0 that is homeomorphic to U_1 by p. Define $\alpha_1 : [0, s_1] \to V_1$ by defining $\alpha_1(s)$ to be the unique point in V_1 with $(p \circ \alpha_1)(s) = \gamma(s)$. Then $\alpha(0) = e_0$ and $p \circ \gamma = \gamma$. Continue this by replacing $\alpha(0)$ with $\alpha(s_1)$ and U_1 with U_2 to extend α_1 to the interval $[0, s_2]$. After m iterations, we arrive at a path $\alpha : [0,1] \to E$, which is unique by the previous lemma, satisfying $\alpha(0) = e_0$ and $p \circ \alpha = \gamma$. ■

Theorem 3.35. *Let $p : E \to X$ be a covering map, and $F : [0,1] \times [0,1] \to X$ be a continuous map, and let $e_0 \in E$ such that $p(e_0) = F(0,0)$. Then there is a unique lift $G : [0,1] \times [0,1] \to E$ of F such that $G((0,0)) = e_0$.*

Proof. Since $[0,1] \times [0,1]$ is connected, uniqueness follows from existence. By the previous theorem, there is a unique path $t \mapsto e_t$ such that $p(e_t) = F(0,t)$ for all $0 \leq t \leq 1$. Similarly, for each fixed t, there is a unique path $s \mapsto G(s,t)$ such that $G(0,t) = e_t$ and $p(G(s,t)) = F(s,t)$ for each s. We need to show that G is continuous. Let $\alpha(s) = G(s,0)$ be the lift of $\gamma(s) = F(s,0)$. We use the same notation as in the previous proof. Then since $\gamma^{-1}(U_j)$ is an open neighborhood of $[s_{j-1}, s_j]$, $F^{-1}(U_j)$ contains $[s_{j-1}, s_j] \times [0, \varepsilon]$ for some small $\varepsilon > 0$. As there are only finitely many U_j's, there is $\varepsilon > 0$ such that

$$F([s_{j-1}, s_j] \times [0, \varepsilon]) \subseteq U_j \quad 1 \leq j \leq m.$$

Now, $G = \left(p|_{V_1}\right)^{-1} \circ F$ on $[0, s_1] \times [0, \varepsilon]$. In particular, G is continuous on $[0, s_1] \times [0, \varepsilon]$. Repeating, G is continuous on $[s_{j-1}, s_j] \times [0, \varepsilon]$ for each j, and hence G is continuous on $[0,1] \times [0, \varepsilon]$. Arguing the same way, for each $t_0 \in (0,1]$, there is $\varepsilon > 0$ such that G is continuous on $[0,1] \times [t_0 - \varepsilon, t_0 + \varepsilon]$. Thus, G is continuous on $[0,1] \times [0,1]$. ■

Remark 3.36. Suppose that γ_1 and γ are loops, and suppose that $\gamma_1 \simeq_{\{0,1\}} \gamma$, and $\{\gamma_t\}_{0 \leq t \leq 1}$ is a homotopy. If $F(s,t) = \gamma_t(s)$, then we can lift the homotopy F to a continuous map $G : [0,1] \times [0,1] \to E$ with $G(0,0) = e$ (for some fixed e such that $p(e) = b = \gamma(0)$) and $p(G(s,t)) = F(s,t) = \gamma_t(s)$. Note that the path $\alpha_t : [0,1] \to E$ given by $\alpha_t(s) = G(s,t)$ is a lift of γ_t and $\alpha_0 = \alpha$. The map $t \mapsto G(0,t)$ is the unique lift starting at e of the constant path at b, and hence the constant path at e. Moreover, $e = G(0,t) = \alpha_t(0)$ for all t, meaning that all α_t's begin at the same point. Similarly, the map $t \mapsto G(1,t)$ is a unique lift starting at $\alpha(1)$ of the constant path at b, which means that the α_t's all end at the same point. In particular, the lift α_1 of γ_1 starts at e and ends at $\alpha(1)$. That is, the ending point of the unique lift (starting at e) of any loop homotopic to γ is the same point $\alpha(1) \in p^{-1}(b)$. This gives us a well defined map

$$\begin{aligned} \phi : \pi_1(X, b) &\to p^{-1}(b) \\ [\gamma] &\mapsto \alpha(1), \text{ where } \alpha \text{ is the lift of } \gamma \text{ starting at } e. \end{aligned}$$

Theorem 3.37. *Let $p : (E, e) \to (X, b)$ be a covering map, and suppose that E is simply connected. Then ϕ as defined in the above remark is a bijection between $\pi_1(X, b)$ and $p^{-1}(b)$.*

Proof. Since E is path connected, the map is surjective: Fix $y \in p^{-1}(b)$ and let α be a path from e to y, and define $\gamma = p \circ \alpha$ then α is the lift of γ starting at e, and $\phi([\gamma]) = \alpha(1) = y$. Now suppose γ_0 and γ_1 are loops in X at b with $\phi([\gamma_0]) = \phi([\gamma_1])$. Take α_0, α_1 to be the lifts of γ_0 and γ_1 at e. Since $\alpha_0(1) = \phi([\gamma_0]) = \phi([\gamma_1]) = \alpha_1(1)$, α_0 and α_1 end at the same point. Thus, $\alpha_0 * \overline{\alpha_1}$ is a loop in E at e. Since E is simply connected, there is a homotopy $F : [0, 1] \times [0, 1] \to E$ from $\alpha_0 * \overline{\alpha_1}$ to the constant loop at e. Then, $p \circ F : [0, 1] \times [0, 1] \to X$ is a homotopy from $\gamma_0 * \overline{\gamma_1}$ to the constant loop at b. That is,

$$[\gamma_0 * \overline{\gamma_1}] = [\gamma_0][\gamma_1]^{-1} = [C_b] \implies [\gamma_0] = [\gamma_1].$$

■

Corollary 3.38. *Let $p : (E, e) \to (X, b)$ be a covering map, and suppose that E is simply connected. For each $y \in p^{-1}(b)$, let α_y be a path in E from e to y, and let $\gamma_y = p \circ \alpha_y$ be a loop in X at b. If γ is any loop in X based at b, then there is a unique loop γ_y such that $[\gamma] = [\gamma_y]$.*

Proof. Let $y = \alpha(1)$, where α is the lift of γ starting at e. Then γ and γ_y are both mapped to y under ϕ, so $[\gamma] = [\gamma_y]$. Hence, such a γ_y exists. Now, if $[\gamma_y] = [\gamma_z]$, then $y = \phi([\gamma_y]) = \phi([\gamma_z]) = z$, so γ_y is unique. ■

Definition 3.39. Let γ be a loop in $\mathbb{S}^1$ based at 1. The unique lift of γ based at 0 in $\mathbb{R}$, $h : [0, 1] \to \mathbb{R}$ such that $e^{2\pi i h(s)} = p \circ h(s) = \gamma(s)$ necessarily has $h(1) \in p^{-1}(1) = \mathbb{Z}$. We call this number the index of γ, denoted $\operatorname{Ind}(\gamma)$.

Theorem 3.40. *The correspondence*

$$\begin{aligned} \pi_1(\mathbb{S}^1, 1) &\to \mathbb{Z} \\ [\alpha] &\mapsto \operatorname{Ind}(\alpha) \end{aligned}$$

is a group isomorphism.

Proof. By the definition of $\operatorname{Ind}(\alpha)$, this map is bijective, so we only need to show it is a homomorphism. This is the same as showing that

$$\operatorname{Ind}(\alpha_1 * \alpha_2) = \operatorname{Ind}(\alpha_1) + \operatorname{Ind}(\alpha_2),$$

where α_1 and α_2 are loops based at 1 in $\mathbb{S}^1$. Take lifts $h_1, h_2 : [0,1] \to \mathbb{R}$ based at $0 \in \mathbb{R}$ of α_1 and α_2 respectively. Then, $\alpha_j(s) = e^{2\pi i h_j(s)}$ for $j = 1, 2$. Define

$$h(s) = \begin{cases} h_1(2s) & 0 \le s \le \frac{1}{2} \\ h_1(1) + h_2(2s-1) & \frac{1}{2} \le s \le 1 \end{cases}.$$

Then $h : [0,1] \to \mathbb{R}$ is continuous and is based at $0 \in \mathbb{R}$. Further,

$$(\alpha_1 * \alpha_2)(s) = \begin{cases} e^{2\pi i h_1(2s)} & 0 \le s \le \frac{1}{2} \\ e^{2\pi i h_2(2s-1)} & \frac{1}{2} \le s \le 1 \end{cases} = \begin{cases} e^{2\pi i h_1(2s)} & 0 \le s \le \frac{1}{2} \\ e^{2\pi i (h_2(2s-1)+h_1(1))} & \frac{1}{2} \le s \le 1 \end{cases}$$
$$= e^{2\pi i \begin{cases} h_1(2s) & 0 \le s \le \frac{1}{2} \\ h_1(1) + h_2(2s-1) & \frac{1}{2} \le s \le 1 \end{cases}} = e^{2\pi i h(s)}.$$

Since h is continuous, it is the unique lift of $\alpha_1 * \alpha_2$, and

$$\mathrm{Ind}(\alpha_1 * \alpha_2) = h(1) = h_1(1) + h_2(1) = \mathrm{Ind}(\alpha_1) + \mathrm{Ind}(\alpha_2),$$

as desired. ■

3.6 Applications of the Index

Notation. We write $\mathbb{B}^2 \subseteq \mathbb{C}$ for the closed unit disc, which has boundary $\mathbb{S}^1$.

Theorem 3.41. *Let f be a continuous map from $\mathbb{B}^2$ to $\mathbb{S}$ that satisfies $f(1) = 1$. Then the loop α in $\mathbb{S}^1$ given by $\alpha(s) = f(e^{2\pi i s})$ has index zero.*

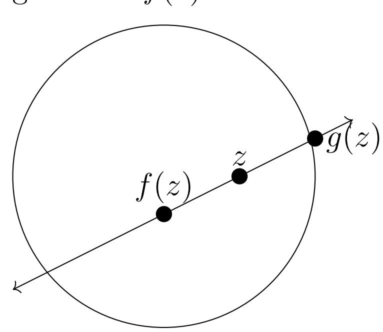
Proof. Let β be the boundary loop of $\mathbb{B}^2$ parameterized by $\beta(s) = e^{2\pi i}$. Then $\alpha = f \circ \beta$. Consider the map given by $F(s,t) = f(\beta(s))^t$. Then F is clearly continuous. Further, $F(s,0) = 1$ is the constant loop in $\mathbb{S}^1$ and $F(s,1) = \alpha(s)$. We also have that $F(0,t) = F(1,t) = 1$, so F is a homotopy between α and the constant loop at 1. As $\mathrm{Ind}(C_1) = 0$, $\mathrm{Ind}(\alpha) = 0$. ■

Corollary 3.42. *There is no continuous map $f : \mathbb{B}^2 \to \mathbb{S}^1$ such that $f(z) = z$ for all $z \in \mathbb{S}^1$.*

Proof. If there were such a map f, then $f(1) = 1$, so the loop $\alpha(s) = f(e^{2\pi i s}) = e^{2\pi i s}$ must have index 0. However, $h : [0,1] \to \mathbb{R}$ given by $h(t) = t$ is a lift of α, so $\mathrm{Ind}(\alpha) = 1$. ■

Theorem 3.43 (Brouwer Fixed Point Theorem in 2 Dimensions)**.** *Any continuous map $f : \mathbb{B}^2 \to \mathbb{B}^2$ has a fixed point.*

Proof. Suppose f has no fixed point. For each $z \in \mathbb{B}^2$, let $g(z)$ be the point of intersection of $\mathbb{S}^1$ and the line passing through z and $f(z)$ that is closer z, as shown below.



We claim g is continuous. Take an open interval $I = \{e^{2\pi it} : a < t < b\}$ in $\mathbb{S}^1$. Assume that $b - a < 2\pi$, as otherwise this is trivial. Suppose that $g(z) \in I$, and take $\varepsilon > 0$ to be small. Take δ_0 so that, for $|z - z_0| < \delta_0$, $|f(z) - f(z_0)| < \varepsilon$. Take δ_1 small enough that, for $|z - y| < \delta_1$ and $|f(z) - w| < \varepsilon$, the line passing through y and w intersects $\mathbb{S}^1$ in I (drawing this out, it is clear that this is possible if we take ε sufficiently small). Set $\delta = \min\{\delta_0, \delta_1\}$. Then, for $|z - w| < \delta$, $g(w) \in I$, so $g^{-1}(I)$ is open. Since g clearly fixes $\mathbb{S}^1$, we have found a continuous map $g : \mathbb{B}^2 \to \mathbb{S}^1$ such that $g(z) = z$ for all $z \in \mathbb{S}^1$, a contradiction. Thus, f must have a fixed point. ■

Theorem 3.44. *Let f be a continuous map from $\mathbb{S}^2$ to $\mathbb{R}^2$. Then there exist antipodal points w and $-w$ in $\mathbb{S}^2$ such that $f(w) = f(-w)$.*

Proof. In this proof, we use the plane $\mathbb{R}^2$ and the complex numbers $\mathbb{C}$ interchangeably when needed. As they have the same topology, aspects of the proof that hold for $\mathbb{R}^2$ also hold for $\mathbb{C}$, and vice versa. Define $g : \mathbb{S}^2 \to \mathbb{R}^2$ by $g(w) = f(w) - f(-w)$. Notice that $g(-w) = -g(w)$. It suffices to show that g vanishes at some point on $\mathbb{S}^2$. Define $h : \mathbb{B}^2 \to \mathbb{R}^2$ by

$$h(x, y) = g(x, y, \sqrt{1 - x^2 - y^2}).$$

Note that h agrees with g on the $\mathbb{S}^1$ that bounds $\mathbb{B}^2$. In particular, for $z \in \mathbb{S}^1$, $h(-z) = -h(z)$. We show that any map with this property must vanish at some point on $\mathbb{B}^2$. Suppose that h does not vanish, and define $\varphi : \mathbb{B}^2 \to \mathbb{S}^1$ by

$$\varphi(z) = \frac{h(z)}{|h(z)|} \cdot \frac{|h(1)|}{h(1)}.$$

Then $\varphi(1) = 1$ and $\varphi(-z) = -\varphi(z)$ for $z \in \mathbb{S}^1$. We now construct a loop α based at 1 by

$$\alpha(s) = \varphi(e^{2\pi is}).$$

Then $\mathrm{Ind}(\alpha) = 0$. We show that $\mathrm{Ind}(\alpha)$ must be odd. Let $k : [0, 1] \to \mathbb{R}$ be the lift of α based at 0, so that $k(0) = 0$ and $\alpha(s) = e^{2\pi ik(s)}$. Then $\mathrm{Ind}(\alpha) = k(1)$. Since $\alpha(s) = \varphi(e^{2\pi is})$, we have that $\varphi(e^{2\pi is}) = e^{2\pi ik(s)}$. Let $z = e^{2\pi is}$ for $s \in [0, 1/2]$. Since $\varphi(-z) = -\varphi(z)$, we have

$$e^{2\pi i(k(s)+1/2)} = -e^{2\pi ik(s)} = -\varphi(z) = \varphi(-z) = \varphi(-e^{2\pi is}) = \varphi(e^{2\pi is}e^{\pi i}) = \varphi(e^{2\pi i(s+1/2)}) = e^{2\pi ik(s+1/2)}.$$

Now,

$$k\left(s + \frac{1}{2}\right) - \left(k(s) + \frac{1}{2}\right) \in \mathbb{Z}.$$

Moreover, since the map from $[0, 1]$ to $\mathbb{R}$ given by

$$s \mapsto k\left(s + \frac{1}{2}\right) - \left(k(s) + \frac{1}{2}\right)$$

is continuous, it must be constant, say equal to $m \in \mathbb{Z}$. Then,

$$k\left(s + \frac{1}{2}\right) - k(s) = m + \frac{1}{2}.$$

Letting $s = \frac{1}{2}$, we have $k(1) - k(1/2) = m + 1/2$, and letting $s = 0$, we have $k(1/2) - k(0) = m + 1/2$. Thus, $k(1) - k(0) = 2m + 1$. Since $k(0) = 0$, $\mathrm{Ind}(\alpha) = k(1) = 2m + 1$, a contradiction. ■

Theorem 3.45 (Ham Sandwich Theorem)**.** *Let U, V and W be bounded, connected and open subsets of $\mathbb{R}^3$. Then there is a plane in $\mathbb{R}^3$ that divides each of the sets into two pieces of equal volume.*

Proof. Assume that all the subsets are nonempty. Given a vector w on the sphere, denote by $h(tw)$ the volume of $U \cap B_{tw}$, where B_{tw} is the side of the plane P_{tw} containing the origin, where P_{tw} is perpendicular to w and passes through tw. If $L_w = \{tw : t \in \mathbb{R}\}$, then the function $h : L_w \to \mathbb{R}$ is continuous, and has range $[0, \mathrm{vol}(U)]$. Let t_w be the (unique, as U is connected) value such that $h(t_w w) = \mathrm{vol}(U)/2$, and let $g_0 : \mathbb{S}^2 \to \mathbb{R}$ be defined by $g_0(w) = t_w$. This is clearly continuous. We also have that $g_0(-w) = -g_0(w)$. Similarly, define maps g_1 and g_2 for V and W. Finally, define $f : \mathbb{S}^2 \to \mathbb{R}^2$ by

$$f(w) = \big(g_0(w) - g_1(w), g_0(w) - g_2(w)\big).$$

By the Borsuk-Ulam theorem, there is $w_0 \in \mathbb{S}^2$ such that $f(-w_0) = f(w_0)$. Now, we have

$$f(-w) = \big(g_0(-w) - g_1(-w), g_0(-w) - g_2(-w)\big) = -\big(g_0(w) - g_1(w), g_0(w) - g_2(w)\big) = -f(w),$$

so $f(w_0) = (0, 0)$. Thus, $g_0(w_0) - g_1(w_0) = 0$ and $g_0(w_0) - g_2(w_0) = 0$, so $g_0(w_0) = g_1(w_0) = g_2(w_0)$. Hence, the plane perpendicular to w_0 passing through $g_0(w_0)w_0$ simultaneously divides the volumes of U, V and W in half. ■